

An EEG study of the combined effects of top-down and bottom-up attentional selection under varying task difficulty

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Abstract

We examined the effect of combined top-down and bottom-up attentional control sources, using known attention-related EEG components that are thought to reflect target selection (N2pc) and distractor suppression (P_D). We used endogenous cues (valid vs. neutral) for top-down attentional control, and salience in the form of color singletons (either the target or a distractor) for bottom-up attentional control in visual search. Crucially, in two experiments, the task was of increasing difficulty, reporting the orientation of a tilted target (Experiment 1), or the position of a small gap within the target among tilted non-targets (Experiment 2). Our results showed strong cueing effects on RT and accuracy in both experiments, demonstrating a general facilitation of responses to validly cued targets. Whereas the processing of salient targets was not improved compared with non-salient targets, the presence of a salient distractor consistently worsened performance. The N2pc and P_D were only observed in trials where targets were preceded by neutral cues in Experiment 1, and for validly cued targets and salient neutrally cued targets in Experiment 2. A cueing effect was found on the P_D in Experiment 1, showing an amplitude reduction in trials where the target was validly cued. These results support the idea that bottom-up attentional allocation occurs only when top-down allocation of attention is absent or inefficient. Therefore, these results indicate that attentional selection and suppression during visual search are both influenced by top-down cueing and give support to theories that focus on the interaction between the two types of attention.

KEYWORDS

attention control, distractor suppression, endogenous cue, N2pc, P_D , salience

1 | INTRODUCTION

Our surroundings provide an overwhelming influx of sensory information that we need to navigate through. For example, traffic-light changes, a turning-light of an approaching car, or a small child running toward a

pedestrian crossing, are just a few signals that a driver needs to encode and process while driving. Some of those signals will be attended and encoded automatically, while others will require the voluntary allocation of attention to be processed. Traditionally, theories of attentional control are divided into bottom-up capture, assuming attention

is attracted by a salient feature of the object without consideration of the observer's goal (e.g., Franconeri & Simons, 2003; Theeuwes, 2010; Yantis & Jonides, 1984), or top-down goal-driven allocation of attention (e.g., Egeth & Yantis, 1997; Folk et al., 1992). This dichotomy, however, has been challenged by findings supporting both accounts, indicating that multiple factors simultaneously contribute to the assignment of processing priorities (e.g., Anderson, 2016; Chelazzi et al., 2013; Gaspelin et al., 2016; Jiang et al., 2015; Jiang et al., 2015; Schapiro et al., 2017; Zelinsky & Bisley, 2015).

1.1 | The interaction between top-down and bottom-up attention-control sources

A well-known paradigm that has often been employed to investigate the effect of different attentional control signals and the interaction between them is visual search, in which the observer needs to search for and discriminate a target embedded in an array of non-targets (Treisman & Gelade, 1980). The target usually has a defining feature to guide the observers in their search, thus, guiding attentional resources in a top-down fashion (e.g., the target is a square among non-targets of different shapes). To manipulate bottom-up allocation of attention, a distractor in the array can be presented as a singleton on the same or on another feature dimension (e.g., the only green among red items). Through such manipulations, it has been demonstrated that responses to the target become slower when a salient distractor is present in the array, suggesting attention is captured by the salient distractor before it can be allocated to the target (e.g., Theeuwes, 1991, 1992). Another powerful manipulation of attention is through spatial cueing (e.g., Posner, 1980). Recently, several studies have examined the interaction between attentional control sources with a cueing paradigm, where attention was captured by a bottom-up exogenous cue, such as a singleton or abrupt onset of a placeholder (e.g., Gaspelin et al., 2016; Su et al., 2020), or guided top-down by endogenous cues where a centrally located symbol directs attention to a spatial location in the visual field (e.g., Berggren & Eimer, 2020; Seiss et al., 2009). As expected, the tendency was that performance improved when the location of the target was cued compared with when it was not. Interestingly, however, salience has been found to be ineffective in capturing attention when there was a mismatch between the observer's goal and the salience-inducing feature (Bacon & Egeth, 1994; Folk et al., 1992; Leber & Egeth, 2006). Folk et al. (1992) for example, showed that an exogenous cue preceding the search array improved performance at the cued location (i.e., faster responses

were found) when the cue was a singleton presented in the same color as the target to be searched, but no such improvement was found for abrupt onset cues, suggesting that whether salience captures attention depends also on the top-down goal directed intentions of the observer in the search task. The present study aims to further explore the interaction between top-down guidance of attention via cues and the allocation of attention through salience, by means of EEG.

1.2 | EEG markers of selection and suppression

Research on attentional control markers in EEG has been thriving (for an overview, see e.g., Eimer, 2014; Luck et al., 2000). In particular, the focus has been on the N2pc component, which is an observed enhanced negativity over posterior electrodes on the contralateral side to the attended stimulus compared with the ipsilateral side, peaking around 200–300 ms from stimulus onset (e.g., Eimer, 1996; Hickey et al., 2008; Hopf et al., 2000; Luck & Hillyard, 1994). The N2pc is thought to reflect attentional selection (e.g., Burra & Kerzel, 2013; Eimer, 1996; Luck & Hillyard, 1994), and possibly more detailed processing at the attended location (e.g., Luck & Hillyard, 1994; Zivony et al., 2018), and it has been observed for both targets and salient distractors, suggesting that the capture of attention is possible by both (Berggren & Eimer, 2020; Hickey et al., 2006; Seiss et al., 2009; Su et al., 2020). However, it was also found that some manipulations of top-down attentional control resulted in the attenuation or elimination of the N2pc related to salience (e.g., Berggren & Eimer, 2020; Burra & Kerzel, 2013; Sawaki et al., 2012; Su et al., 2020). For example, Berggren and Eimer (2020) showed that the N2pc for targets in a go/no-go task that appeared on the side of the display that should be ignored was attenuated, indicating effective spatial filtering of salient irrelevant items.

The *signal suppression hypothesis* by Luck and colleagues (Gaspelin & Luck, 2018a, 2018b; Sawaki & Luck, 2010) posits that salience can capture attention unless top-down suppression is imposed. They propose an inhibitory mechanism, which is directed at suppressing salient irrelevant stimuli that would capture attention otherwise, acting before attention shifts to that stimulus location. In support of this hypothesis, salient distractors have been found to elicit a posterior-contralateral positivity, known as P_D , which is believed to reflect the active suppression of that item (e.g., Berggren & Eimer, 2020; Burra & Kerzel, 2013; Gaspelin & Luck, 2018b; Liesefeld et al., 2017; Sawaki et al., 2012). The time window in which the P_D can arise is wide, sometimes following the N2pc and sometimes occurring during, or before it, indicating different

stages of distractor inhibition (Berggren & Eimer, 2020). A recent study by Liesefeld et al. (2017; see also Feldmann-Wüstefeld et al., 2021, and Liesefeld et al., 2021) tracked the temporal dynamics of attentional shifts during visual search in the presence of a salient distractor as reflected by the N2pc and P_D . That study provided a direct comparison of target and distractor N2pc and P_D by comparing conditions in which the target was presented laterally while the salient distractor was presented on the midline or vice versa, thereby, measuring lateral activity for each of them individually (see Hickey et al., 2006). Liesefeld et al. (2017) found that distractor N2pc was observed at an earlier time window compared with target N2pc, preceding the P_D . This pattern indicated that attention was captured by the salient distractor before the target, and then redirected toward the target while the distractor was actively suppressed. Similar findings of attentional shifting were also observed recently with event-related magnetic field recordings (Donohue et al., 2018). Still, others have demonstrated that attentional capture indicated by an N2pc to the distractor was likely to emerge when the target and the distractors were relatively similar to each other, leading to an increased task difficulty, while distractor suppression indicated by a P_D was likely to emerge in easier tasks, where target-distractor similarity was low (Barras & Kerzel, 2017). Thus, whether suppression or capture are observed seems to depend on task difficulty. Similar observations have been made recently with single-neuron recordings and EEG in monkeys, showing a differentiation in prefrontal neurons in response to salient distractors that suggested that the selection of the salient item by these neurons was not automatic, and was subjected to task demands (Cosman et al., 2018). Similar conclusions were obtained in a recent study using machine-learning decoding on prefrontal cortical activity in monkeys (Di Bello et al., 2020).

1.3 | The logic of the current study

Previous studies on attentional control, and specifically related EEG markers, vary significantly in their use of paradigm and task. Common top-down control manipulations have been focused on changing the attentional set of the observer by instructions leading to a feature or singleton search mode (e.g., Burra & Kerzel, 2013; Gaspelin & Luck, 2018a, 2018b; Kiss et al., 2012, 2013; Su et al., 2020), or the allocation of attention to a certain area of the display while ignoring the rest (e.g., Berggren & Eimer, 2020; Sawaki et al., 2012). Several studies used a spatial cue to allocate attention to the target location. Su et al. (2020) and Huang et al. (2016) used a color singleton cue that preceded a search array of either a color or shape

singleton target. The cue could appear at the prospective target location or at a different location, rendering it valid in the former and invalid in the latter case. They found that N2pc for invalidly cued targets was attenuated, but still present in both cases. Notably, Huang et al. (2016) also found no N2pc for shape singleton targets that followed a color singleton cue unrelatedly to its validity, signifying the intricate relationship between top-down and bottom-up signals of attentional control. Still, only a few studies employed a cue that carried only spatial information and was irrelevant to the target feature to be searched (Berggren & Eimer, 2020; Kiss et al., 2008; Seiss et al., 2009). Such central cues promote a volitional allocation of attention to the target location on every trial without changing the attentional set of the observer. Furthermore, a few studies used neutral cues rather than invalid ones. For example, Kiss et al. (2008) found similar N2pc amplitudes to shape singleton targets in visual search, following both informative and uninformative cues about the side of the display that the target would appear in. Similarly, Seiss et al. (2009), using a go/no go task, showed that N2pc for color targets was elicited on both cued and non-cued sides, suggesting bottom-up capture of attention by the salient feature despite the instructions to ignore the targets in the non-cued side of the display.

The current study aimed to systematically investigate the relationship between salience and endogenous cueing of top-down attention via central cues in visual search, both for target selection and distractor suppression. Specifically, we were interested in the combined effect of bottom-up salience of either the target or the distractor and top-down endogenous cueing on the N2pc and P_D . We measured these components using a display in which for any given trial, only the target or the salient distractor was lateralized, in order to purely characterize these lateralized components (Hickey et al., 2006). We manipulated target salience, the presence of a salient distractor, and whether spatial information about the target location was provided on each trial through a cue that was either valid or neutral. Since task difficulty has been shown to affect the emergence of the N2pc and P_D (e.g., Barras & Kerzel, 2017; Luck et al., 1997), we manipulated this factor in two experiments. In Experiment 1 the task was relatively easy, as participants reported the orientation of the target that was an oblique singleton among horizontal and vertical non-targets, resulting in low target-nontargets similarity and a relatively easy discrimination task. In Experiment 2, the participants reported the position of a gap within the oblique target that was surrounded by non-targets that were also tilted, and thus more similar to the target, both of which increase task difficulty. In both experiments, the search-array was preceded by a central cue that pointed at the prospective target location (valid) or did not carry

spatial information (neutral). In addition to being a singleton regarding its orientation, the target was sometimes a color singleton (red among green items or vice versa); on other trials, one of the non-targets was the color singleton.

We expected that performance would generally improve on trials with a valid cue. As it has been demonstrated previously that in top-down cueing tasks the global orientation of attention to a pre-cued location is reflected by a contingent negative variation (CNV), and its amplitude has been linked to how strongly attention is engaged (e.g., Schevernels et al., 2014; van den Berg et al., 2014). Hence, we expected that the general improvement in performance following a valid cue might be accompanied by an enhancement of the CNV. With regard to the N2pc, several predictions could be made; one is that a comparable N2pc would be elicited for targets following both types of cues, similarly to the findings of Kiss et al. (2008). However, if our valid cue to the target location enables efficient top-down allocation of attention in advance of the target onset, the N2pc might be attenuated for targets following a valid cue, as limited further processing of the target would be required once it is found (Kiss et al., 2008; Luck & Hillyard, 1994; Tan & Wyble, 2015).

Regarding the interaction between top-down and bottom-up attentional control sources, if top-down attentional control prevents bottom-up allocation of attention due to salience as part of a filtering mechanism, no benefit should be observed on performance for salient targets following a valid cue. Similarly, the presence of a salient distractor in the search array should have a detrimental effect on performance after a neutral cue but not after a valid one. Such gating should also be reflected in the N2pc. Specifically, we expected that target salience would influence the N2pc following a neutral cue, and that this effect would be attenuated for salient targets following a valid cue. Moreover, we expected that the N2pc might show a differentiation between the different search-display configurations after a neutral cue, in particular that the N2pc would be larger in the presence of a salient distractor, as the N2pc has been found to be enhanced in the presence of distractors that leads to an increased difficulty in discrimination (e.g., Boehler et al., 2011; Hopf et al., 2002). Importantly, no such differentiation was expected after valid cues.

Turning to distractor processing, according to the signal suppression hypothesis, top-down suppression will not be activated if attention is already strongly focused elsewhere (e.g., Sawaki & Luck, 2010). Thus, we hypothesized that the P_D would be elicited more strongly (or even exclusively) by a salient distractor following a neutral cue, reflecting active suppression. As task difficulty was manipulated between the two experiments, we hypothesized that different patterns of the N2pc and P_D may emerge in

the different experiments. In particular, the salient distractor may elicit a P_D in Experiment 1 and an N2pc in Experiment 2.

2 | EXPERIMENT 1

2.1 | Method

2.1.1 | Participants

Twenty-five volunteers participated in Experiment 1, of which four were excluded from further analysis due to technical issues. The remaining 21 participants (1 male, age $M = 23.43$ ($SD = 3.94$), 1 left-handed) were naïve to the purpose of the study and were reimbursed for participation. The sample size was based on typical sample sizes of studies in the field and participants' availability. All had normal or corrected-to-normal vision and reported no neurological deficits. Informed consent was obtained before the experiment, and the protocol was approved by the ethical committee of the Faculty of Psychology and Educational Sciences of Ghent University (code 2018/13).

2.1.2 | Stimuli and design

Stimuli were presented using PsychoPy (v1.84.2) software (Peirce, 2007) on a 24in. Benq XL2411Z LED monitor controlled by a Dell Optiplex 9020 tower with Intel Core i5-4590 processor, at 60-Hz refresh-rate. To monitor eye movements, we used a 250Hz SMI eye tracker (RED250 mobile system; SensoMotoric Instruments, Teltow, Germany). Participants rested their heads on a cushioned chinrest at a viewing distance of 60 cm. Examples of the different search displays are depicted in Figure 1. Search displays consisted of four $2^\circ \times 0.5^\circ$, green (RGB: 0, 86, 0; 138.5 cd/m²) and red (RGB: 170, 0, 0; 64.8 cd/m²) bars on a gray background (RGB: 128, 128, 128; 85.5 cd/m²). The bars were positioned in a diamond-like arrangement centered 7° away from a white fixation cross ($0.5^\circ \times 0.5^\circ$; RGB: 255, 255, 255; 190.2 cd/m²). The target was tilted 20° clockwise or counterclockwise from vertical. The three remaining bars (i.e., non-targets) were oriented vertically or horizontally with a constraint that not all three had the same orientation. Overall, there were four conditions, three aimed at measuring the N2pc and one for measuring the P_D . To measure the N2pc, the target appeared on the left or right location in the search display. In the *no-distractor* condition, all four bars were in the same color (i.e., green or red). In the *salient-target* condition, the target had a different color than the three non-targets (i.e., red target and green non-targets or vice versa). In the *lateral-target*

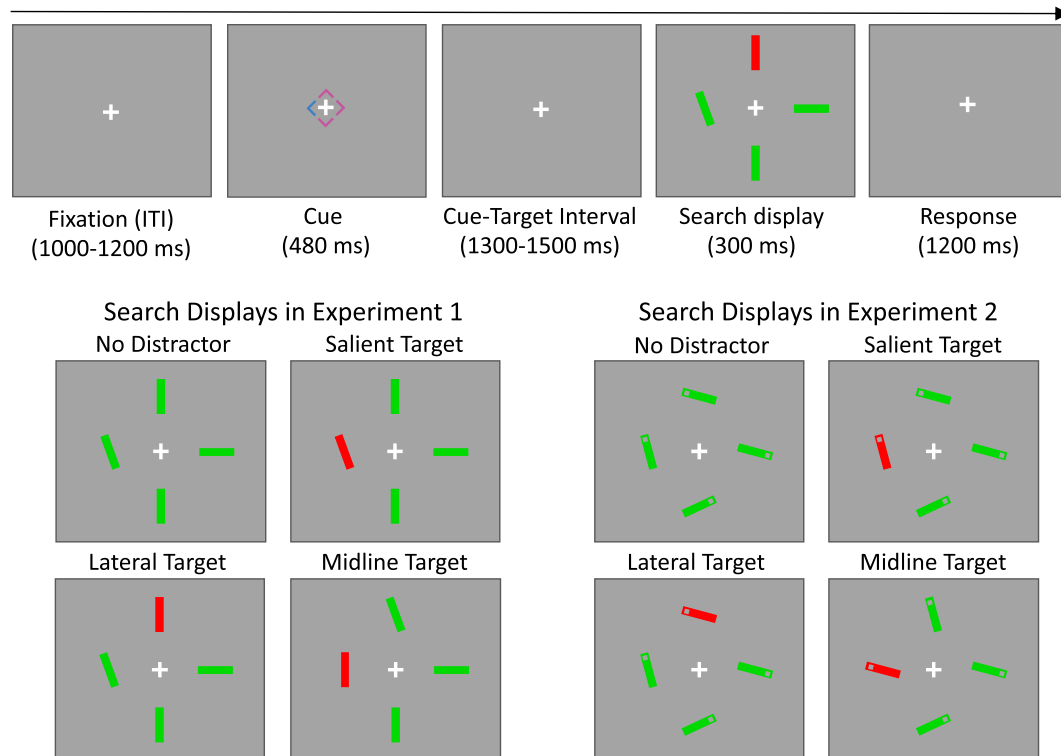


FIGURE 1 Top: sequence of events in a trial in Experiment 1, depicting a valid cue to the location left of fixation in a lateral-target condition. Bottom: four search-display conditions in Experiment 1 (left) and Experiment 2 (right). In the *no-distractor* condition the target appeared laterally without a salient distractor. In the *salient-target* condition, the target appeared laterally and was the only salient item in the display. In the *lateral-target* condition, the target appeared laterally while a salient distractor appeared at the top or bottom location. In the *midline-target* condition, the target appeared at the top or bottom location, while the salient distractor appeared laterally. In both experiments the target was the only item tilted right or left from vertical. The task in Experiment 1 was to report the orientation of the tilt, whereas in Experiment 2 the task was to report the location of the gap within the target (up or down). In both experiments, on half of the trials the cardinal color was green (salient target and distractors were red), while on the other half of the trials the colors were reversed

condition one of the non-targets was of a different color while the other two had the same color as the target (i.e., a red distractor in a green display or a green distractor in a red display). Importantly, the salient distractor appeared in the upper or lower location so that the measured lateral activity (i.e., N2pc) reflected only target processing. Counter to this, in order to measure the P_D , in the *midline-target* condition, the salient distractor appeared on the left or right location, while the target appeared at the upper or lower location. On half of the displays the principal color was green, while on the other half it was red. Each condition was presented equally often. Thus, the targets appeared lateralized on 75% of all trials (25% in each of the three condition) and 25% of the trials it appeared on the vertical midline.

Cues were composed of L-shape segments forming a $1.2^\circ \times 1.2^\circ$ diamond around the fixation cross. A neutral cue was composed of four magenta segments (RGB: 120, 0, 90; 89.5 cd/m^2). For valid cues, one of the segments was cyan (RGB: 0, 56, 158; 81.1 cd/m^2), pointing to the location of the prospective target. A neutral cue appeared on half of the trials and a valid cue on the other half. Due to the

specific arrangement of the target and non-targets in the search display, with only one of the four search-display conditions with a target on the vertical meridian, cues to the left or right locations appeared on 75% of valid-cue trials (37.5% each), while in the other 25% the cue pointed to the upper or lower locations (12.5% each).

2.1.3 | Procedure

All the conditions were presented in a randomized order. The sequence of events in a trial is depicted at the top part of Figure 1. A fixation cross was presented continuously at the center of the screen. After a cue presentation of 480 ms and a cue-target interval (CTI) of 1300–1500 ms, the search display was presented for 300 ms. The participants' task was to search for the tilted target and indicate its orientation (clockwise or counterclockwise rotation from vertical) with a keypress using the index fingers of the left and right hand ("m" for right tilt and "z" for left). Responses were recorded from onset of the search display until 1200 ms after display offset, for a total of 1500 ms.



Once the time-window for response had passed, an inter-trial interval displaying only the fixation cross was jittered between 1000 and 1200 ms. Participants were instructed to fixate on the fixation cross that was presented continuously, and to not move their eyes. Fixation was monitored continuously by the experimenter by means of the online eye-position display of the eye-tracker, and the participants were warned in case eye-movements were observed to allow correction. Due to technical issues with the eye-tracker, central fixation could not be directly monitored for two of the participants but was clearly instructed. Participants performed 10 trials as practice and 12 experimental blocks of 96 trials (1152 trials in total). Thereby, each condition of cue and display combination was presented for 144 trials.

2.1.4 | Electrophysiological recording and analysis

EEG data were recorded using a Brain Products actiCHamp 64-channel system (Brain Products, Gilching, Germany) with 64 active scalp electrodes positioned according to the standard international 10–10 system. The EEG data were analyzed using Brain Vision Analyzer 2 (Brainproducts, Gilching, Germany). Signals were recorded at a 500-Hz sampling rate with Fz as the reference during recording. During the analysis the data were re-referenced offline to the average of TP9 and TP10, which correspond to the left and right mastoids, and Fz was restored to the data set. Segments of the raw EEG data that showed clearly identifiable artifacts (not including blinks and eye movements) were excluded during manual inspection, and a high-pass filter of 0.1 Hz was applied. Next, independent component analysis (ICA) was used to remove components related to eyeblinks from the data. Eye-movements were observed very rarely, and respective ICs were removed from the data. The data were limited to trials with correct responses only and were segmented into epochs from –200 to 4000 ms relative to cue onset, and baseline-corrected with respect to the 200-ms pre-cue period. A semi-automatic artifact rejection blind to the experimental conditions was performed on these segments. This operation led to the exclusion of about 6% of the trials on average (range: 0.3%–24%). For analyzing the data elicited by the search arrays, the data were segmented further into epochs from –200 to 800 ms relative to search-display onset, according to the different cues, search display conditions, and target or distractor location (left vs. right), and baseline corrected relative to the 200-ms pre-search-display period. Then, data in the different conditions were averaged per participant. Epochs were initially averaged separately for right and left target or distractor locations

for each participant in each condition. Then, epochs were averaged for contralateral and ipsilateral channels according to the appropriate location. Lastly, difference waves were calculated for each participant in each condition by subtracting their average ipsilateral from their average contralateral signal. Statistical analyses reported in the result section were performed on these difference waves.

We quantified three components of interest (CNV, N2pc, P_D ; see below). In each case, we followed the collapsed-localizer approach (Luck & Gaspelin, 2017) by collapsing across all conditions, and determining both the electrodes and time-ranges of interest based on this average signal.¹ For the CNV, this could be done for the simple average across all valid and neutral cues, then selecting the maximally negative channel, and at this channel the time-range displaying a gradually increasing negativity from its start until the end of the CTI. For the N2pc and P_D , this selection was done with the corresponding contralateral-minus-ipsilateral difference waves, collapsed across the search-display conditions with a lateral target (no-distractor, lateral-target, salient-target) and cues (neutral and valid) for the N2pc, and across cues (neutral and valid) for the P_D , identifying the most negative/positive electrode in the respective time-ranges, which were again also selected based on the average difference-wave signal as a transient relative negativity (for the N2pc) and positivity (for the P_D). Based on this approach, we selected the signal at Cz between 700 and 1780 ms after cue onset to quantify the CNV. The end point of the CNV analysis was chosen as the earliest point in which the search display could appear. The N2pc and P_D were both found to be maximal at PO7/PO8, with the relevant time-ranges being 202–314 ms for the N2pc, and 314–418 ms for the P_D . Because the collapsed localizer approach may bias toward the detection of statistically significant differences from zero, we ran a permutation test (e.g., Maris & Oostenveld, 2007), to evaluate the probability of achieving significance for the N2pc and P_D components (i.e., respective negative or positive activity significantly different from zero) in the different conditions. Details of this procedure are provided as Supporting Information.

¹An additional advantage of this approach is that it is entirely data-driven, which further helped us in compensating for a small mistake during recording. Specifically, due to a measurement error in Experiment 1, the cap was placed further to the back, positioning FCz exactly at the Cz site. At the sites that are traditionally considered for the N2pc and P_D , which are usually PO7 and PO8, the channel pair that was the closest (~1 cm off) was P3 and P4, which was also found to display the largest N2pc and P_D here. For consistency, the results reported here refer to the correct sites on the head for the two experiments, although they appear under different names in the recorded data.

All the statistical analyses reported in the article were performed in R (R Core Team, 2016) with ez (Lawrence, 2011/2015) and effectsize (Ben-Shachar et al., 2020) packages. The permutation analyses were performed with Python version 3.9.7 and the NumPy toolbox (Harris et al., 2020). p values were corrected with Greenhouse-Geisser epsilon in cases of significant sphericity violation. Planned comparisons for confirmation of hypotheses are reported with no correction to the p value. Analyses of RTs as well as the CNV, N2pc and P_D included trials with correct responses only.

2.2 | Results

2.2.1 | Behavior

Figure 2 shows the results for accuracy (AC) and RTs in the different conditions of Experiment 1. Accuracy and RT data were subjected to separate 2×4 rm-ANOVAs, with cue (valid, neutral) and search display condition (lateral target, no-distractor, salient target, midline target) as within-subject factors. A significant main effect of cue was found for accuracy and RT [AC: $F(1,20) = 16.26$, $p = .0007$, $\eta_p^2 = 0.49$; RT: $F(1,20) = 64.85$, $p < .0001$, $\eta_p^2 = 0.76$], as well as a significant main effect of search display condition [AC: $F(3,60) = 3.84$, $p = .01$, $\eta_p^2 = 0.84$; RT: $F(3,60) = 37.90$, $p < .0001$, $\eta_p^2 = 0.65$]. These effects were qualified by a significant interaction both for accuracy and RT [AC: $F(3,60) = 3.32$, $p = .03$, $\eta_p^2 = 0.14$; RT: $F(3,60) = 25.87$, $p < .0001$, $\eta_p^2 = 0.56$, respectively]. An inspection of the

accuracy interaction suggested that it was mostly driven by the attenuation of the cueing effect when the targets were salient. For RT, the interaction seemed to be driven by a larger cueing benefit when a salient distractor was present in the search display, and in particular when it was lateralized. This could have been due to the target appearing more frequently in the lateralized positions, resulting in more interference from the salient distractor when a neutral cue preceded the target that appeared at a less frequent location.

To test the hypothesis that salient targets may have a benefit over non-salient targets that would possibly be attenuated following a valid cue, we conducted a separate 2×2 rm-ANOVA with cue (valid, neutral) and search-display, removing the conditions containing a salient distractor (i.e., keeping the salient-target and no-distractor conditions). This test revealed significant main effects of cue [AC: $F(1,20) = 8.04$, $p = .01$, $\eta_p^2 = 0.29$; RT: $F(1,20) = 46.03$, $p < .0001$, $\eta_p^2 = 0.70$], search display for RT [AC: $F < 1$; RT: $F(1,20) = 5.11$, $p = .04$, $\eta_p^2 = 0.20$], and an interaction for accuracy [AC: $F(1,20) = 9.85$, $p = .005$, $\eta_p^2 = 0.33$; RT: $F < 1$]. Paired t -tests (two-tailed) showed a marginally significant effect demonstrating higher accuracy for salient targets compared with non-salient ones following a neutral cue [$t(20) = 1.99$, $p = .06$, Cohen's $d = 0.43$], but the effect was reversed following a valid cue [$t(20) = -1.79$, $p = .08$, Cohen's $d = -0.39$]. To test the hypothesis that cueing effect will attenuate the interference from a salient distractor another 2×2 rm-ANOVA was conducted for cue and search display with salient distractor absent/present. For this, search-display

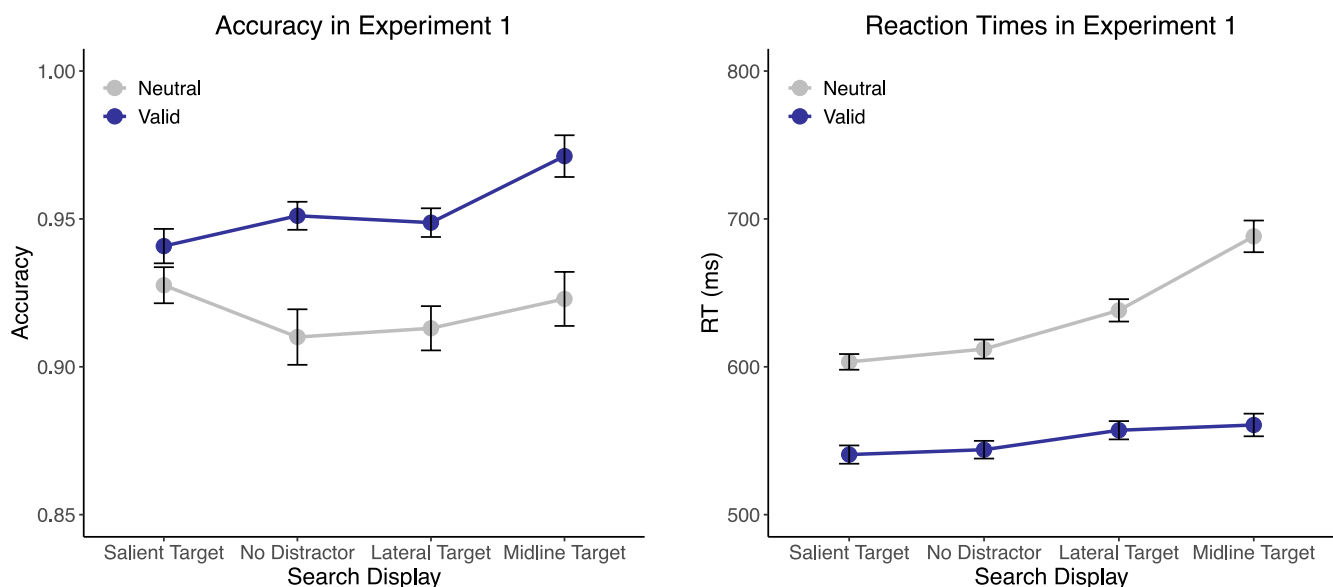


FIGURE 2 Mean accuracies (left) and reaction times (right) as a function of cue and search display conditions in Experiment 1. Error bars indicate within-subject standard errors as suggested by Cousineau (2005) with the correction suggested by Morey (2008)

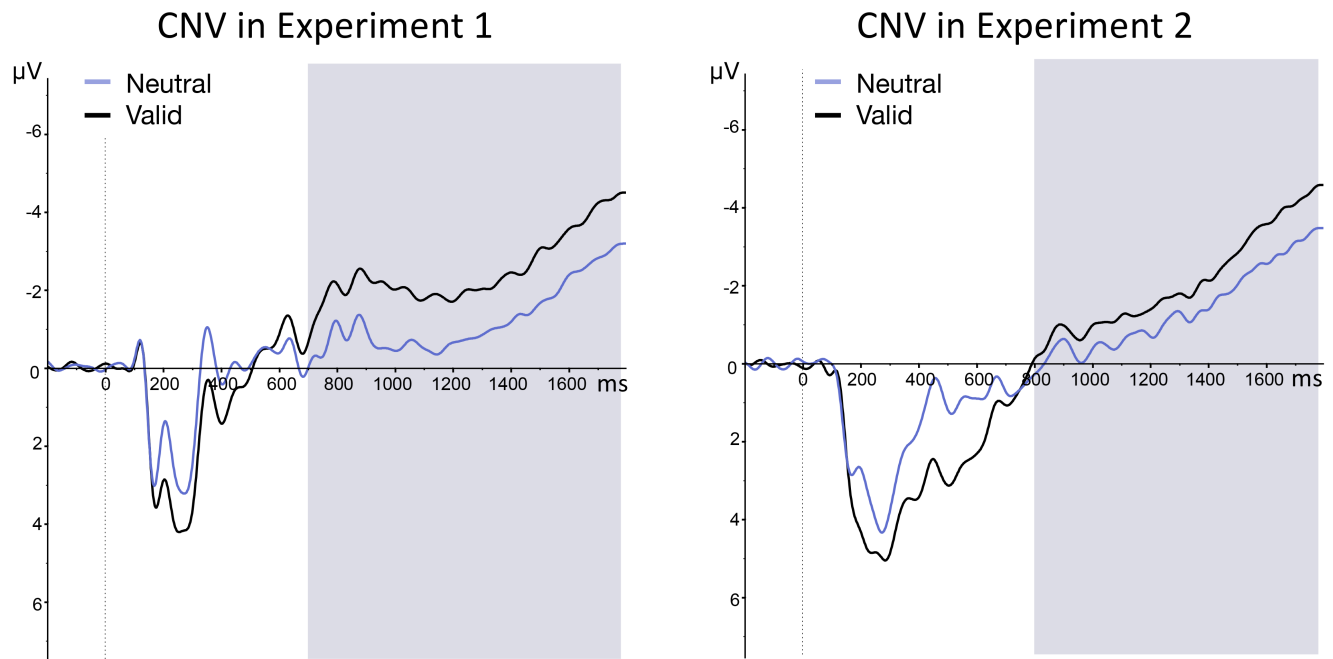


FIGURE 3 ERP waves following valid and neutral cue conditions in Experiment 1 (left) and 2 (right), showing a cueing effect on the contingent negative variation (CNV). Time-point zero indicates the cue onset. In both experiments, the end point of the CNV analysis shows as the earliest point in which the search display could appear

conditions where a salient distractor was present (i.e., lateral target, midline target) were collapsed,² and compared to the no-distractor condition. This revealed a significant main effect of cue on accuracy and RT [AC: $F(1,20) = 19.23$, $p = .0003$, $\eta_p^2 = 0.49$; RT: $F(1,20) = 62.30$, $p < .0001$, $\eta_p^2 = 0.77$], as well as a significant main effect of distractor condition [AC: $F(1,20) = 3.28$, $p = .09$, $\eta_p^2 = 0.14$; RT: $F(1,20) = 76.41$, $p < .0001$, $\eta_p^2 = 0.79$], and an interaction between the two factors on RT [AC: $F < 1$; RT: $F(1,20) = 33.21$, $p < .0001$, $\eta_p^2 = 0.62$]. Paired t -tests (two-tailed) revealed that RTs were significantly slower in the salient-distractor-present conditions following both valid [$t(20) = 3.32$, $p = .003$, Cohen's $d = 0.72$], and neutral cues [$t(20) = 7.83$, $p < .0001$, Cohen's $d = 1.71$], thus, an impairment caused by the salient distractor was evident in the both conditions but it was larger in the neutral ($M = 51$ ms) than in the valid ($M = 15$ ms) condition.

²Separate 2×2 ANOVAs conducted on the non-collapsed data showed a similar pattern: lateral target versus no distractor—main effect of cue [AC: $F(1,20) = 15.34$, $p = .0008$, $\eta_p^2 = 0.44$; RT: $F(1,20) = 47.53$, $p < .0001$, $\eta_p^2 = 0.70$], main effect of distractor condition [AC: $F < 1$; RT: $F(1,20) = 46.63$, $p < .0001$, $\eta_p^2 = 0.70$], and the interaction between the two factors [AC: $F < 1$; RT: $F(1,20) = 4.98$, $p = .04$, $\eta_p^2 = 0.20$]; midline target versus no distractor—main effect of cue [AC: $F(1,20) = 19.09$, $p = .0003$, $\eta_p^2 = 0.49$; RT: $F(1,20) = 73.41$, $p < .0001$, $\eta_p^2 = 0.79$], main effect of distractor condition [AC: $F(1,20) = 6.51$, $p = .02$, $\eta_p^2 = 0.23$; RT: $F(1,20) = 41.50$, $p < .0001$, $\eta_p^2 = 0.68$], and the interaction between the two factors [AC: $F < 1$; RT: $F(1,20) = 32.97$, $p < .0001$, $\eta_p^2 = 0.62$].

2.2.2 | Contingent negative variation

A CNV was found in both conditions separately, as indicated by amplitudes significantly below zero following both valid [$t(20) = -6.32$, $p < .0001$, Cohen's $d = -1.38$] and neutral cues [$t(20) = -4.02$, $p = .0003$, Cohen's $d = -0.88$]. To test whether cueing the target location had an effect on preparatory processes following the cue, a paired t -test (two-tailed) was performed, revealing increased negativity following valid ($M = -2.48$) compared with neutral ($M = -1.21$) cues [$t(20) = -5.29$, $p < .0001$, Cohen's $d = -1.15$] (Figure 3). This effect indicates that the cues were utilized in preparation for the target display, and more so when the target location was predicted.

2.2.3 | N2pc

Turning to the search displays, Figure 4 displays the data collapsed across hemisphere (i.e., left and right) into activity contralateral and ipsilateral to the item of interest (target or a salient distractor), as well as difference waves and the time interval used to quantify the two amplitudes of the components in Experiment 1. To check whether the component was elicited in each of the conditions, one-sample t -tests (one-tailed) showed a significant N2pc (mean amplitude lesser than zero) for targets following a neutral cue [lateral-target: $t(20) = -4.05$, $p = .0001$, Cohen's $d = -0.98$;

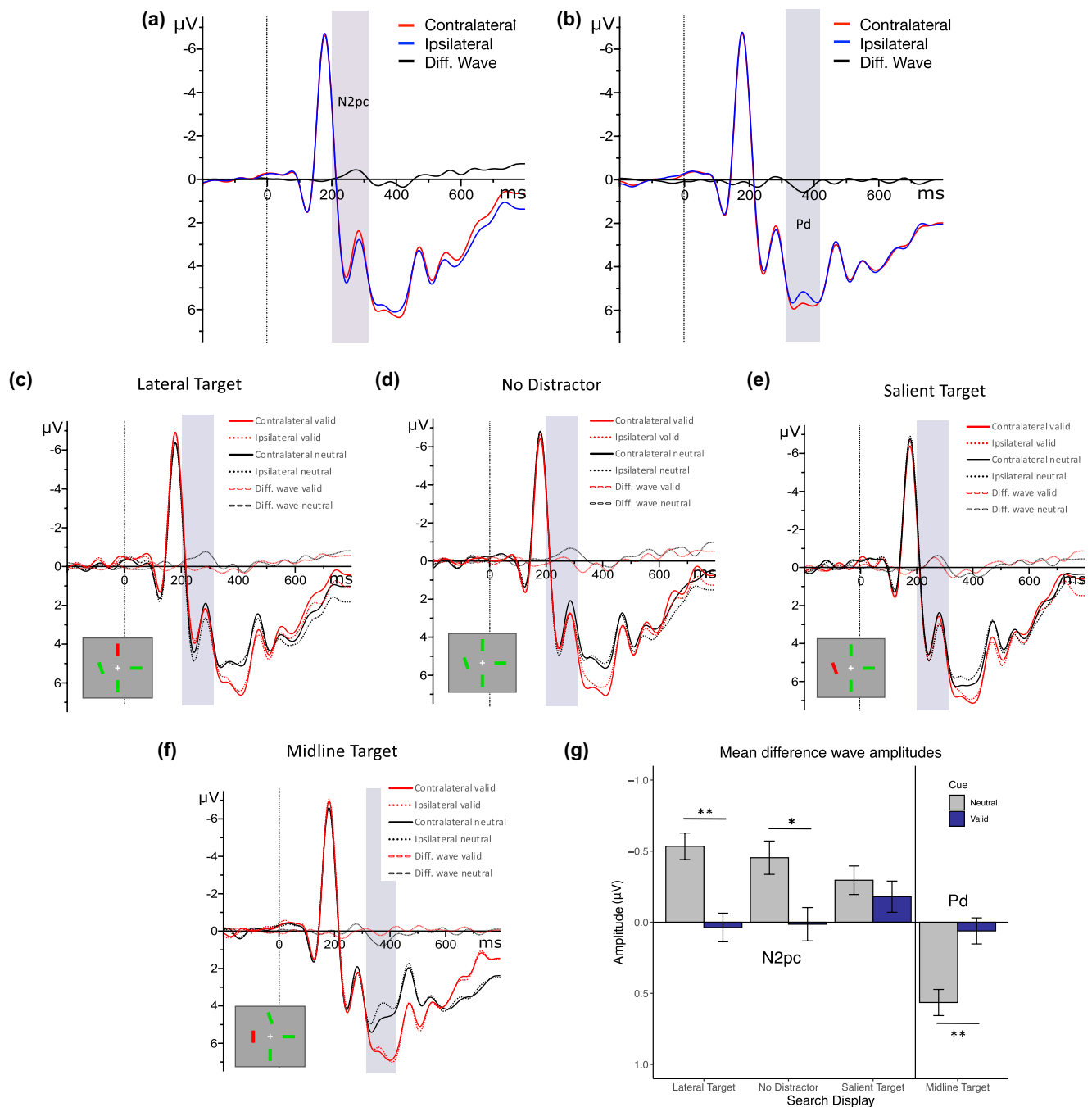


FIGURE 4 Contra/Ipsilateral and difference wave ERPs in Experiment 1. Time-point zero indicates the search-display onset. Shading represents mean-amplitude time window for N2pc and P_D components at electrode sites PO7/8 for the condition depicting a lateralized target (a) and midline-target condition (b). Time windows and electrodes of interest were determined using collapsed localizers (Luck & Gaspelin, 2017), see text for details. Cueing effects on N2pc (c–e) and P_D (f) in the different search display conditions. Mean difference waves amplitudes in the different cue and search display conditions for N2pc and P_D (g). Error bars indicate within-subject standard errors as suggested by Cousineau (2005) with the correction suggested by Morey (2008)

no-distractor: $t(20) = -3.16$, $p = .003$, Cohen's $d = -0.69$; salient-target: $t(20) = -2.09$, $p = .03$, Cohen's $d = -0.46$], but not following a valid cue [lateral-target: $t(20) = 0.28$, $p = .61$, Cohen's $d = 0.04$; no-distractor $t(20) = -0.08$, $p = .53$, Cohen's $d = -0.02$; salient target: $t(20) = -1.17$, $p = .13$, Cohen's $d = -0.26$]. The permutation tests confirmed these results (Figure S1d–f). This pattern was

further confirmed by a 2×3 rm-ANOVA with cue (valid, neutral) and search display (lateral-target, no-distractor, salient-target) conducted on the mean amplitude of N2pc difference waves from the three conditions depicting a lateral target as within-subject factors. The analysis revealed a significant main effect of cue [$F(1,20) = 6.26$, $p = .02$, $\eta_p^2 = 0.24$], and an interaction between the two factors

$[F(2,40) = 4.02, p = .03, \eta_p^2 = 0.17]$. No significant effect of search display was found $[F < 1]$.

Planned comparisons (paired t -test, two-tailed) revealed that the cueing effect on N2pc was significant in the lateral-target condition $[0.57 \mu\text{V}, t(20) = 3.62, p = .002, \text{Cohen's } d = 0.79]$, and in the no-distractor condition $[0.47 \mu\text{V}, t(20) = 2.28, p = .03, \text{Cohen's } d = 0.50]$, but not significant in the salient-target condition $[0.12 \mu\text{V}, t(20) = 0.64, p = .53, \text{Cohen's } d = 0.14]$. This pattern suggests that the interaction between cue and search display relates to a reduced cueing effect in the salient-target compared with the other conditions. To test the specific hypothesis that the cueing effect on the N2pc is attenuated for salient targets as the former t -tests suggested, we conducted a separate 2×2 rm-ANOVA with cue (valid, neutral) and search-display (salient-target, no-distractor). This analysis revealed a marginally significant interaction between the two factors $[F(1,20) = 3.36, p = .08, \eta_p^2 = 0.14]$. To test the hypothesis that the N2pc is enhanced in the presence of a salient distractor, and that this effect would possibly be attenuated following a valid cue, another 2×2 rm-ANOVA was conducted with the factors cue (valid, neutral) and search display (lateral-target, no-distractor). This test revealed only a significant main effect of cue $[F(1,20) = 9.41, p = .006, \eta_p^2 = 0.32]$, and no other effect was significant $[F_s < 1]$. Thus, the interaction observed earlier was indeed due to the different magnitudes of cueing effects between the salient-target and the other two display conditions. Our results generally suggest that once attention was allocated to the target location by the cue, no additional attention that is reflected by the N2pc was needed in that location for the target to be processed. While there was a significant interaction between cueing and search display, no clear effects could be found for target salience and the presence of a salient distractor.

2.2.4 | P_D

Lastly, we investigated the presence and modulations of the P_D component by focusing on the midline-target condition. One-sample t -tests (one-tailed) showed a significant P_D (mean amplitude greater than zero) in neutrally $[t(20) = 3.13, p = .003, \text{Cohen's } d = 0.68]$, but not in validly cued trials $[t(20) = 0.53, p = .30, \text{Cohen's } d = 0.12]$. Again, these results were confirmed by the permutation tests (Figure S1g,h). To test for cueing effects on the P_D , mean amplitudes of the difference waves from the midline-target condition were subjected to a two-tailed paired t -test. As predicted, the analysis showed a significant effect of cue $[0.56 \mu\text{V}, t(20) = 3.16, p = .005, \text{Cohen's } d = 0.69]$, indicating a reduced P_D amplitude following a valid cue to the target location. This result suggests that

the distractor suppression reflected in the P_D is tightly correlated with the allocation of attention by the endogenous cue. Presumably, when attention is efficiently allocated to a different area, no suppression is activated because it is not needed. This interpretation is in accordance with the signal-suppression hypothesis (e.g., Gaspelin & Luck, 2018c) that argues that suppression is activated for irrelevant salient objects that fail to capture attention. However, because the P_D reported here occurred in a time-window that is less likely to reflect salience detection (e.g., Feldmann-Wüstefeld et al., 2021), it is possible that active suppression was simply not needed following a valid cue, irrelevant of whether a salient distractor was present.

2.3 | Discussion

Our hypotheses were generally confirmed, showing a large behavioral cueing effect per se, as well as a general differentiation across search-display conditions. These observations were qualified by significant interactions suggesting some benefit for target salience following a neutral cue, but no benefit following a valid cue. Also, an interference from a salient distractor was observed following both valid and neutral cues, however, this interference was attenuated by top-down guidance of attention.

The lack of an N2pc for validly cued targets in Experiment 1 does not accord with findings of other studies that used cues to predict the target location (e.g., Berggren & Eimer, 2020; Seiss et al., 2009; Su et al., 2020). However, it accords with studies showing that the N2pc is not elicited for certain tasks, where finding and reporting the target can be done relatively easily (Kiss et al., 2008; Luck et al., 1997; Luck & Hillyard, 1994). Luck et al. (1997), for example, showed a reduction in N2pc when participants performed a simple feature search task, compared with a more difficult discrimination task. In the current search-display setting, discriminating the target from the non-targets could be done easily due to the target being the only oblique bar in the display, thus, rendering it very dissimilar to the other bars and causing it to pop out. In addition, no further processing of the target was required once it was found, as the task was to report the orientation of the target, which was also the feature guiding the search. Presumably, attentional operations that are reflected in the N2pc were no longer required for the task to be completed once attention was already shifted to the target location (Tan & Wyble, 2015). This is furthermore in line with the lack of significant differences between search displays; a salient target did not trigger an increased N2pc, presumably because its salience was not used for guiding attention to the target and processing it, since the participants were engaging in feature search (Barras & Kerzel,

2016, 2017; Luck et al., 1997), here, in the orientation dimension. Similarly, the presence of a salient distractor did not lead to an increased N2pc because the target was already very easy to select, so the salient distractor did not lead to greater competition (Barras & Kerzel, 2016).

Taken together, the absence of a clear N2pc and P_D following a valid cue suggests that top-down allocation of attention via a central pre-cue facilitated target selection to an extent that rendered the processes underlying the N2pc and P_D unnecessary once the target appeared, which furthermore went along with a reduced need for suppression of a salient distractor. As mentioned in the introduction, since the task was relatively easy to perform, and the presence of the N2pc and P_D has been found to vary depending on task difficulty (e.g., Barras & Kerzel, 2017; Luck et al., 1997), we were interested to see if these results would be replicated in a comparable setting where the task was more demanding. Hence, we conducted Experiment 2 to test the same general hypotheses specified in the introduction with a more difficult task, with some additional expectations. To that end, the search array contained non-targets that were also tilted away from the cardinal axes, making them more similar to the target and, thus, leading to more competition (Barras & Kerzel, 2017). Instead of reporting the target tilt, participants had to indicate the location of a gap at one of the target's ends. If additional attention is needed for target processing in more difficult tasks despite allocation of attention to the target location in advance, an N2pc should be elicited for targets following a valid cue in Experiment 2. Still, the N2pc amplitude may be attenuated following a valid compared with a neutral cue, as in the latter more attention could be required. On the other hand, the increased similarity between the target and the other items in the array could result in reduced N2pc amplitudes following a neutral cue, if the target is less often selected instantaneously. In addition, the P_D has been shown to be elicited for salient distractors in easy tasks, where target-non-targets similarity was low (Barras & Kerzel, 2017). Thus, P_D is expected to be reduced compared to Experiment 1.

3 | EXPERIMENT 2

3.1 | Method

3.1.1 | Participants

Twenty-three volunteers participated in Experiment 2, of whom two were excluded from further analysis due to technical issues and another three due to low performance accuracy ($\leq 60\%$). The remaining 18 participants (2 men, age $M = 22.44$ ($SD = 3.11$), 3 left-handed) were naïve to the purpose of the study and were paid for participation.

All had normal or corrected-to-normal vision and reported no neurological deficits. Informed consent was obtained before the experiment, and the protocol was approved by the ethical committee of the Faculty of Psychology and Educational Sciences of Ghent University (code 2018/13).

3.1.2 | Stimuli, design, procedure, and EEG pre-processing

Experiment 2 was identical to Experiment 1 except for the following: The target was tilted 25° clockwise or counterclockwise from vertical. The three non-targets were tilted 25° clockwise or counterclockwise from horizontal. Each bar had a “gap” on one side (i.e., a square with the same color as the background). The participants were instructed to search for the target and report on which side of it the gap was located (i.e., “m” for top and “z” for bottom). To reduce the duration of the experiment, the ITI (i.e., fixation time before cue display), which already followed a long post-stimulus time-window (1200 ms following the time-limit for making a response after search-display offset) was shortened to 100–300 ms. Fixation was monitored by an eye-tracker, and a trial was terminated when fixation was broken over a 2° window around fixation. Due to technical issues with the eye-tracker, central fixation could not be used for trial termination for four participants but was clearly instructed and monitored. One of those participants showed excessive eye-movements and was excluded from the analyses.

Very noisy electrodes detected during recording or visual inspection of the raw data were interpolated for four participants (1–2 electrodes each). After the semi-automatic artifact rejection procedure, the average total of excluded segments was 11% (range: 0.1%–23%). Using the collapsed-localizer approach described for Experiment 1, the CNV was quantified at Cz between 800 and 1780 ms from cue onset, and the N2pc for the target at PO7/PO8 between 182 and 296 ms from search display onset. No negativity indicating an N2pc for the distractor was observed, but rather a positivity compatible with the P_D at PO7/PO8 between 298 and 398 ms from search display onset.

3.2 | Results

3.2.1 | Behavior

Figure 5 shows the results for accuracy and RTs in the different conditions of Experiment 2. Accuracy and RT data for correct trials were subjected to separate 2-way rm-ANOVAs, with cue (valid vs. neutral) and search display condition (lateral-target, no-distractor, salient-target,

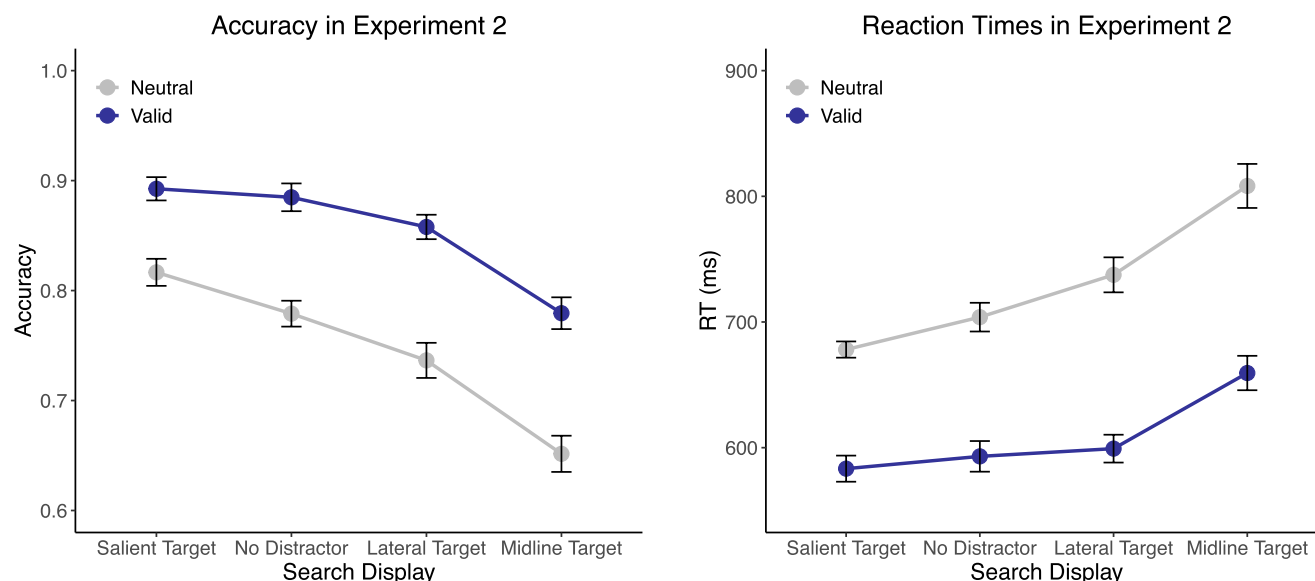


FIGURE 5 Mean accuracies (left) and reaction times (right) as a function of cue and search display conditions in Experiment 2

midline-target), as within-subject factors. Similar to the results of Experiment 1, a significant main effect of cue was found on accuracy and RT [AC: $F(1,16) = 37.62, p < .0001, \eta_p^2 = 0.70$; RT: $F(1,16) = 37.36, p < .0001, \eta_p^2 = 0.70$], as well as significant main effect of search display condition on both measures [AC: $F(3,48) = 69.42, p < .0001, \eta_p^2 = 0.81$; RT: $F(3,48) = 73.82, p < .0001, \eta_p^2 = 0.82$]. These effects were qualified by a significant interaction between the two factors for RT and marginally significant for accuracy [AC: $F(3,48) = 2.64, p = .09, \eta_p^2 = 0.14$; RT: $F(3,48) = 8.58, p = .001, \eta_p^2 = 0.35$]. An inspection of the interactions suggests that similar to Experiment 1, they were mostly driven by the attenuation of the cueing effect when the targets were salient, and a larger cueing benefit when a salient distractor was present in the search display.

To test the hypothesis of a benefit for target salience that would possibly be attenuated following a valid cue, we conducted a separate 2×2 rm-ANOVA with cue (valid, neutral) and search-display, removing the conditions containing a salient distractor (i.e., keeping the salient-target and no-distractor conditions). This test revealed significant main effects of cue [AC: $F(1,16) = 25.44, p = .0001, \eta_p^2 = 0.61$; RT: $F(1,16) = 35.29, p < .0001, \eta_p^2 = 0.69$], and search display [AC: $F(1,16) = 9.34, p = .008, \eta_p^2 = 0.37$; RT: $F(1,16) = 24.23, p = .0002, \eta_p^2 = 0.60$], and a significant interaction for accuracy [AC: $F(1,16) = 4.99, p = .04, \eta_p^2 = 0.24$; RT: $F(1,16) = 2.85, p = .11, \eta_p^2 = 0.15$]. Paired t -tests (two-tailed) showed a significant effect demonstrating higher accuracy for salient targets compared with non-salient ones following a neutral cue [$t(16) = 3.06, p = .008, \text{Cohen's } d = 0.74$], but no effect following a valid cue [$t(16) = 1.12, p = .28, \text{Cohen's } d = 0.27$].

To test the hypothesis that the cueing effect will attenuate the interference from a salient distractor, another 2×2 rm-ANOVA was conducted for cue and search display with salient distractor absent/present. For this, search-display conditions where a salient distractor was present (i.e., lateral-target, midline-target) were collapsed,³ and compared to the no distractor condition. This revealed a significant main effect of cue on accuracy and RT [AC: $F(1,16) = 36.33, p < .0001, \eta_p^2 = 0.69$; RT: $F(1,16) = 33.52, p < .0001, \eta_p^2 = 0.68$], as well as a significant main effect of search display [AC: $F(1,16) = 102.79, p < .0001, \eta_p^2 = 0.87$; RT: $F(1,16) = 78.17, p < .0001, \eta_p^2 = 0.83$], and a significant interaction between the two factors on RT and marginally significant for accuracy [AC: $F(1,16) = 2.24, p = .15, \eta_p^2 = 0.12$; RT: $F(1,16) = 10.67, p = .005, \eta_p^2 = 0.40$]. Paired t -tests (two-tailed) revealed that responses were significantly less accurate and slower in the salient distractor present conditions following both valid [AC: $t(16) = 6.60, p < .0001, \text{Cohen's } d = 1.60$; RT: $t(16) = 5.22, p < .0001$,

³Separate 2×2 ANOVAs conducted on the non-collapsed data showed a similar pattern: lateral target versus no distractor—main effect of cue [AC: $F(1,16) = 28.89, p < .0001, \eta_p^2 = 0.64$; RT: $F(1,16) = 32.87, p < .0001, \eta_p^2 = 0.67$], main effect of distractor condition [AC: $F(1,16) = 17.78, p = .0007, \eta_p^2 = 0.53$; RT: $F(1,16) = 23.41, p = .0002, \eta_p^2 = 0.59$], and an interaction between the two factors [AC: $F(1,16) = 1.91, p = .19, \eta_p^2 = 0.11$; RT: $F(1,16) = 12.96, p = .002, \eta_p^2 = 0.45$]; midline target versus no distractor—main effect of cue [AC: $F(1,16) = 35.93, p < .0001, \eta_p^2 = 0.69$; RT: $F(1,16) = 32.73, p < .0001, \eta_p^2 = 0.67$], main effect of distractor condition [AC: $F(1,16) = 123.49, p < .0001, \eta_p^2 = 0.89$; RT: $F(1,16) = 85.10, p < .0001, \eta_p^2 = 0.84$], and the interaction between the two factors [AC: $F < 1$; RT: $F(1,16) = 6.92, p = .02, \eta_p^2 = 0.30$].

Cohen's $d = 1.27$], and neutral cues [AC: $t(16) = 8.97$, $p < .0001$, Cohen's $d = 2.18$; RT: $t(16) = 8.04$, $p < .0001$, Cohen's $d = 1.95$], thus, as in Experiment 1, an impairment caused by the salient distractor was evident in both conditions, but it was larger in the neutral ($M = 8.5\%$, 66 ms) than in the valid ($M = 6.6\%$, 35 ms) condition.

To examine the effect of task difficulty on the performance in the task, we conducted mixed-ANOVAs on the accuracy and RT data with experiment as between-subjects and cue as within-subject factors. These analyses showed a significant main effect of experiment for accuracy and marginally significant for RT [AC: $F(1,36) = 46.83$, $p < .0001$, $\eta_p^2 = 0.57$; RT: $F(1,36) = 3.86$, $p = .06$, $\eta_p^2 = 0.10$], and a significant effect of cue [AC: $F(1,36) = 59.66$, $p < .0001$, $\eta_p^2 = 0.62$; RT: $F(1,36) = 93.83$, $p < .0001$, $\eta_p^2 = 0.72$], and an interaction between the two factors for accuracy [AC: $F(1,36) = 15.78$, $p = .0003$, $\eta_p^2 = 0.31$; RT: $F(1,36) = 2.83$, $p = .10$, $\eta_p^2 = 0.07$]. These results indicate that performance in Experiment 1 was generally better ($M = 93.6\%$; 593 ms) than in Experiment 2 ($M = 80.0\%$; 664 ms), and that the cueing effect was larger for accuracy in Experiment 2 (10.8%) compared with Experiment 1 (3.5%).

3.2.2 | Contingent negative variation

As in Experiment 1, a CNV was found in both cueing conditions, as indicated by amplitudes significantly below zero following both valid [$t(16) = -4.34$, $p = .0003$, Cohen's $d = -1.05$] and neutral cues [$t(16) = -1.91$, $p = .04$, Cohen's $d = -0.46$]. A paired t -test (one-tailed) was performed, revealing that although numerically there seemed to be an increased negativity following valid ($M = -2.08$), compared with neutral ($M = -1.39$) cues, the difference between the two conditions was not significant [$t(16) = 1.17$, $p = .26$, Cohen's $d = 0.28$] (Figure 3). Mixed-ANOVA conducted on the data from the two experiments with experiment as between-subjects and cue as within-subject factors revealed a significant main effect of cue [$F(1,36) = 4.86$, $p = .03$, $\eta_p^2 = 0.12$], no effect of experiment [$F < 1$], nor an interaction between the two factors [$F(1,36) = 1.21$, $p = .28$, $\eta_p^2 = 0.03$].⁴

⁴An analysis on the CNV data in a time-window of 700–1780 ms, as in Experiment 1, showed similar results; mean amplitude was significantly lower than zero for the valid condition ($t(16) = -3.91$, $p = .0006$, Cohen's $d = -0.95$), and marginally significant for the neutral condition ($t(16) = -1.71$, $p = .05$, Cohen's $d = -0.41$), no significant cueing effect ($t(16) = -1.08$, $p = .30$, Cohen's $d = -0.26$), and only a significant effect of cue in the mixed-ANOVA ($p = .03$; experiment: $F < 1$; interaction: $p = .23$).

3.2.3 | N2pc

Figure 6 displays the data collapsed across hemisphere (i.e., left and right) into activity contralateral and ipsilateral to the item of interest (target or a salient distractor), as well as difference waves and the time interval with maximum amplitude for the two components in Experiment 2. In contrast to the results observed in Experiment 1, an N2pc was now found for targets following a valid cue [lateral-target: $t(16) = -1.84$, $p = .04$, Cohen's $d = -0.44$; no-distractor: $t(16) = -1.73$, $p = .05$, Cohen's $d = -0.42$; salient-target: $t(16) = -1.68$, $p = .06$, Cohen's $d = -0.41$], and only for salient targets following a neutral cue [lateral-target: $t(16) = 0.19$, $p = .57$, Cohen's $d = 0.05$; no-distractor: $t(16) = 0.30$, $p = .62$, Cohen's $d = 0.07$; salient-target: $t(16) = -1.86$, $p = .04$, Cohen's $d = -0.45$]. The permutation tests confirmed these results (Figure S1i–k).

Mean amplitude of difference waves from the three conditions with a lateral target were subjected to a 2×3 rm-ANOVA with cue (valid, neutral) and search display (lateral-target, no-distractor, salient-target) as within-subject factors. The analysis showed only a marginally significant effect of cue (cue: [$F(1,16) = 3.22$, $p = .09$, $\eta_p^2 = 0.17$]; search display: [$F(2, 32) = 1.31$, $p = .29$, $\eta_p^2 = 0.08$]; interaction: [$F < 1$]). Planned comparisons (t -test, two-tailed) showed that the cueing effect on N2pc was either marginally significant or did not reach statistical significance in the three conditions [lateral-target condition: $0.53 \mu V$, $t(16) = 1.84$, $p = .08$, Cohen's $d = 0.45$; no-distractor: $0.52 \mu V$, $t(16) = 1.72$, $p = .10$, Cohen's $d = 0.42$; salient-target: $0.31 \mu V$, $t(16) = 1.02$, $p = .32$, Cohen's $d = 0.25$].

As in Experiment 1, to test the specific hypothesis that the cueing effect on the N2pc will be attenuated for salient targets following a valid cue, we conducted a separate 2×2 rm-ANOVA with cue (valid, neutral) and search-display (salient-target, no-distractor). However, this test revealed no significant effects ($ps > .12$). To test the hypothesis that the N2pc will be enhanced in the presence of a salient distractor, and that this effect will possibly be attenuated following a valid cue, another 2×2 rm-ANOVA was conducted for cue and search display (lateral-target, no-distractor). As in Experiment 1, this test revealed only a marginally significant main effect of cue [$F(1,16) = 4.08$, $p = .06$, $\eta_p^2 = 0.20$], and no other effect was significant [$Fs < 1$]. Mixed-ANOVA conducted on the data from the two experiments with experiment as between-subjects and cue as within-subject factors revealed no main effects [experiment: $F(1,36) = 2.62$, $p = .11$, $\eta_p^2 = 0.07$; cue: $F < 1$], but a significant interaction between cue and experiment [$F(1,36) = 10.61$, $p = .003$, $\eta_p^2 = 0.23$].

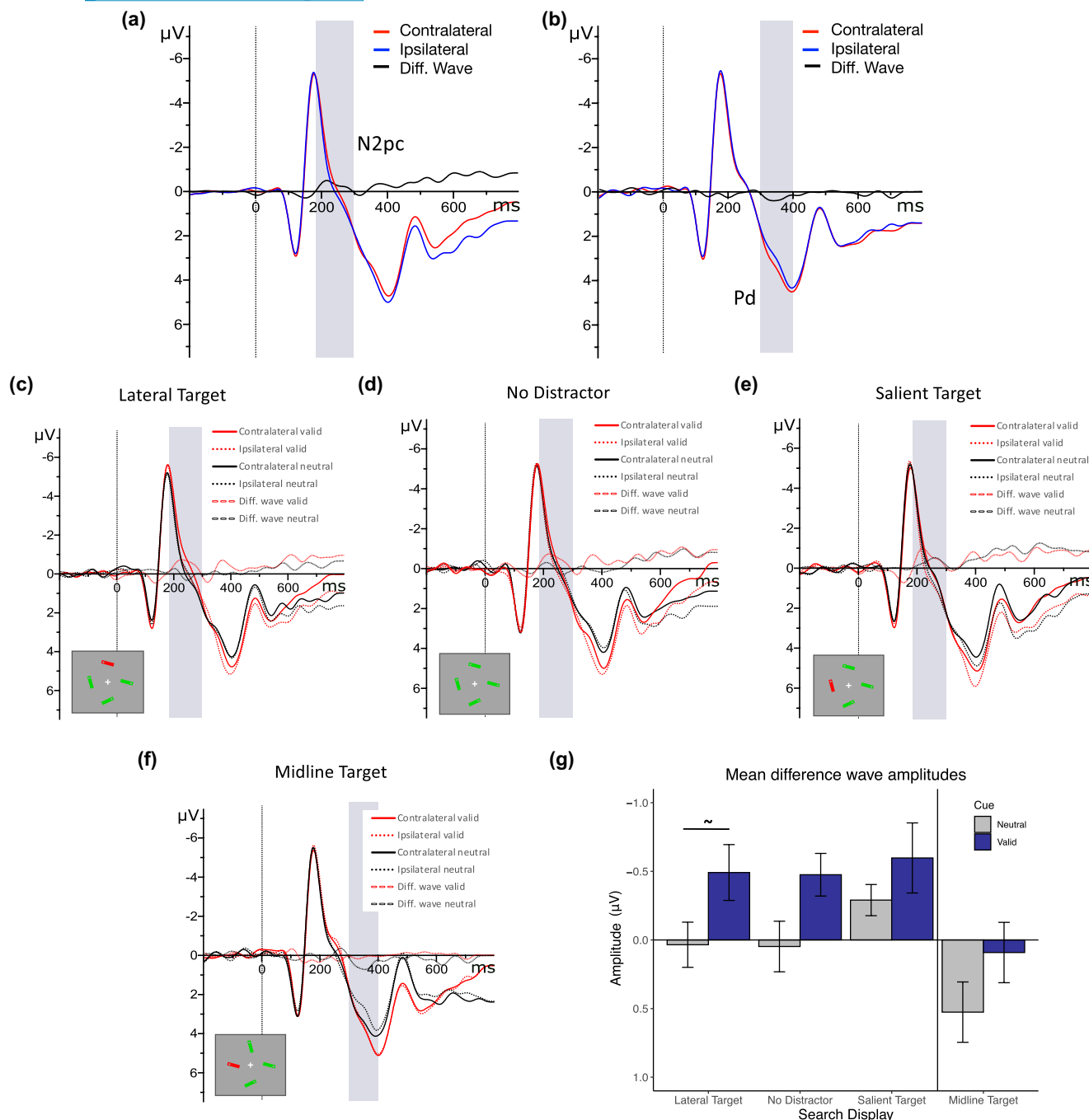


FIGURE 6 Contra/Ipsilateral and difference wave ERPs in Experiment 2. Time-point zero indicates the search-display onset. Shading represents mean-amplitude time windows for N2pc and P_D components at electrode sites PO7/8 for the conditions depicting a lateralized target (a) and midline-target condition (b), averaged across cueing conditions. Time windows were determined using collapsed localizers (Luck & Gaspelin, 2017), see text for details. Cueing effects on N2pc (c–e) and P_D (f) in the different search display conditions. Mean difference waves amplitudes in the different cue and search display conditions for N2pc and P_D (g). Error bars indicate within-subject standard errors as suggested by Cousineau (2005) with the correction suggested by Morey (2008)

3.2.4 | P_D

Looking at the P_D , a significant P_D was observed in the neutral but not in the valid cue condition [$t(16) = 2.44$, $p = .01$, Cohen's $d = 0.59$; $t(16) = 0.37$, $p = .36$, Cohen's $d = 0.09$, respectively]. These results were confirmed by

permutation tests (Figure S10,p). To test for cueing effects on the P_D , mean amplitudes of the difference waves from the midline-target condition were subjected to a two-tailed paired t -test. The analysis showed no significant effect of cue [$0.43 \mu V$, $t(16) = 1.14$, $p = .27$, Cohen's $d = 0.28$]. Mixed-ANOVA conducted on the data from the

two experiments with experiment as between-subjects and cue as within-subject factors revealed a significant main effect of cue [$F(1,36) = 5.94, p = .02, \eta_p^2 = 0.14$], and no main effect of experiment or an interaction [$F_s < 1$].

3.3 | Discussion

The behavioral results of Experiment 2 generally replicate those of Experiment 1, showing facilitation for processing in a validly cued target location in the presence of a salient distractor, as well as an attenuation of the effect of target salience when a valid cue preceded the target. In addition, they confirm the hypothesis that changing the task and display setting would lead to the task being more difficult to perform, and thus, the valid cue could lead to a larger benefit.

Ambiguous evidence was found for the utilization of the cue in the second experiment. The results of experiment 1, showing a cueing effect on the CNV amplitude, were not replicated here. This could be interpreted as the valid cue being used less effectively in the current setting, resulting in weaker preparation for the upcoming target, or that the neutral cue prompted more engagement in preparation for the upcoming array, or possibly both. Still, the results of the comparison between the experiments support the notion that the cues were similarly utilized in the two experiments, likely somewhat stronger after valid cues.

The interaction found between cue and experiment confirms that the cue affected processing at the target location differently in the two experiments; In Experiment 2, the N2pc found for targets following a valid cue, albeit not very robust, suggests that further attentional processing was needed at the target location in order to perform the task in this experiment, even if attention could have been shifted to the target location in advance. In contrast, the absence of a clear N2pc for most targets following a neutral cue is not easily explained. An N2pc was observed for salient targets following a neutral cue, suggesting that in the absence of an efficient top-down allocation of attention, bottom-up allocation of attention was enabled. Because in the other two conditions the target was not salient enough, it was probably often not found instantaneously, hence, no N2pc was observed. The reason for the change in search strategy for the neutrally cued targets is presumably due to high search difficulty resulting from the similarity between the target and distractor (Barras & Kerzel, 2017). It is possible that under these conditions, where no guidance of attention was enabled by a cue or salience, participants resorted to using a serial search, selecting items in a random fashion. Yet, for the N2pc to be observable, it is important that attention

is quickly focused on the target item (or at least the target hemifield). So, such variance in where attention is initially directed would decrease the possibility of observing the N2pc elicited by the target, and potentially the N2pc elicited by the salient distractor in a difficult task (Barras & Kerzel, 2017). An alternative explanation could be that the neutral cue led to diffused attention over the whole search array. This explanation is supported by the N2pc to the salient targets following a neutral cue but challenged by the lack of a clear P_D to the salient distractor.

The results concerning the P_D also suggest that the change of setting affected suppression of distracting salient information, as the increased similarity between the target and its surrounding items increased interference and reduced active suppression (Barras & Kerzel, 2017). This result furthermore accords with the observed N2pc for salient targets following a neutral cue, as bottom-up allocation of attention due to salience was presumably readily employed for target discrimination. This also matches the observed effect of the salient distractor on behavior, showing larger interference following a neutral cue. Still, it is also possible that the switch in search strategy following a neutral cue resulted in the salient distractor being selected at random along with the target, preventing active suppression to be applied, or the P_D from being observed like the N2pc in Experiment 2. Taken together, the results indicate a large difference in how the cue affected the combination of top-down and bottom-up sources of attention in the two experiments.

4 | GENERAL DISCUSSION

This study set out to examine the combined effects of top-down allocation of attention by an endogenous cue and bottom-up allocation of attention by stimulus salience. We did so in a series of two experiments, in which the difficulty of the visual-search task was increased for the second. The EEG recordings were then probed for markers of attentional orienting after the cue (CNV), as well as for putative correlates of target selection (N2pc) and distractor inhibition (P_D). The CNV, a component related to preparatory attentional orienting, was found to have a larger amplitude after a valid cue, mostly in Experiment 1, but was generally present after both cue types in both experiments. During the subsequent search displays, the presence of an N2pc was strongly influenced by whether or not a valid cue was presented in a fashion that differed markedly across the two experiments. Specifically, the N2pc was only clearly observed after neutral cues in Experiment 1, and only after valid cues in Experiment 2, with the exception of salient targets following a neutral cue. In addition, the data indicate that the P_D was larger

after neutral cues, a condition which requires more distractor suppression than after valid cues.

Concerning the combination of bottom-up salience and top-down endogenous attentional control, we had two main hypotheses. Specifically, we expected that a salient target would have little benefit following a valid cue, and that a salient distractor would cause less interference when attention is allocated to the target location in advance compared with when there is no top-down guidance of attention. Supporting the idea that top-down allocation of attention can prevent salience from being useful in a search task (e.g., Sawaki & Luck, 2010), both our hypotheses were supported; following a valid cue, little to no difference in performance was found between salient and non-salient targets, and better performance was observed in the presence of a salient distractor compared with following a neutral cue. Interestingly, no difference in the N2pc amplitude was found between salient and non-salient targets in both experiments, and there was no strong effect of distractor salience (when presented at the midline) on the N2pc.

It should be noted, however, that making inferences from the comparison between the conditions in which the target was lateralized and the condition in which it appeared on the midline are complicated by the imbalance of target locations frequencies, as it has been shown that statistical learning of the target location affects performance in the search (Geng & Behrmann, 2002). Namely, the target appeared laterally on 75% of the trials, while it appeared on the vertical midline for the rest 25%. Of particular interest, in the lateral-target condition the target appeared in the more likely location and the distractor appeared in the less likely location, thus, the search might have been facilitated by statistical learning of target location, also reducing the interference from the distractor as it appeared in a lower priority location. In contrast, in the midline-target condition the imbalance was reversed; the target appeared in the less likely location making the search more difficult, and the distractor appeared in the more likely target location, causing more interference. The behavioral results accord with these predictions, showing worse performance, and a larger cueing effect in the latter case than in the former. Furthermore, the ERPs corresponding to these two conditions may reflect somewhat different processes, which are impossible to distinguish with the present data.

More generally, when comparing the two experiments, the N2pc, which is thought to reflect target selection, was affected differently by the cue manipulation in the two experiments. In Experiment 1, the N2pc was elicited only for targets that followed a neutral cue, suggesting that top-down allocation of attention via the valid cue eliminated

the need for further attentional processing at that location to perform the task. Yet, the opposite was found in Experiment 2, where the N2pc was elicited for validly cued targets but not for neutrally cued ones, except for salient targets. This difference between the two experiments suggests that in Experiment 1 top-down attentional control was sufficient to perform the task, however, that was not the case in Experiment 2, where a more difficult setting was used. Consequently, further attentional engagement for target processing was required in the latter even for targets following a valid cue.

4.1 | Task settings affect target selection

The different N2pc patterns in the two experiments confirmed the hypothesis that the combination of attentional control sources can be affected by task difficulty. In Experiment 1, a valid cue facilitated the search for the target-defining feature, which was also the feature that needed to be discriminated. Thus, target processing was relatively simple and therefore likely did not elicit an N2pc. Following a neutral cue, however, the search was not facilitated, and the selection and discrimination of the target elicited an N2pc. Importantly, compared with Experiment 1, the search in Experiment 2 was more difficult, as evident from the longer RTs and lower accuracy rates. Specifically, the target was more similar to the other items in the array, thus, no longer a salient orientation singleton. Therefore, more scrutiny was required to find and discriminate the uniquely tilted target. The finding that validly cued targets elicited an N2pc in Experiment 2 but not in Experiment 1 is in accord with previous studies that showed larger N2pc amplitudes in difficult search conditions, when the target and non-targets were similar, compared with easier searches, where the target popped out (Luck & Hillyard, 1994). Yet, another difference between the search tasks in our two experiments is the processing steps that needed to occur; In Experiment 1 the participants reported the same target feature they were searching for (i.e., the tilt of the target), but in the task in Experiment 2, after discriminating the target from the non-targets through its tilt, the participants still needed to locate a gap within the target and to report its position. Such tasks were often employed in studies that observed an N2pc that was interpreted as attentional shifting toward the target (e.g., Berggren & Eimer, 2020; Burra & Kerzel, 2013; Liesefeld et al., 2017; but see Kiss et al., 2008). Because we assume that attention was shifted to the target location in advance when a valid cue was provided, the observed N2pc suggests that it reflects successive processing at that location (e.g., Kiss et al., 2008; Theeuwes, 2010; Zivony et al., 2018), whereas the N2pc

elicited in the absence of attentional guidance to the target location may reflect attentional shifting and target processing (e.g., Theeuwes, 2010). A previous study (Töllner et al., 2012), examined the differences between discrimination and compound tasks in visual search, such as the ones used in our study, and showed no effect of task on the N2pc (or PCN as it is sometimes referred to, e.g., Töllner et al., 2012), suggesting that this difference in task might not affect attentional dynamics. However, in that study the search was guided by target salience and no top-down guidance of attention was provided in advance. It should also be noted that two other methodological changes that were introduced in Experiment 2, namely, the shortening of the ITI and the trial termination that followed if participants broke central fixation, may have influenced the participants to adopt a different search strategy compared with Experiment 1. Exactly how these factors might affect the components examined here merits further research.

A recent paper by Liesefeld and Müller (2020) proposes a new theoretical framework of a dichotomy between *priority guidance* and *clump scanning* as search strategies in a visual search task. In priority guidance, priority maps are integrated from salience maps of different feature dimensions and top-down guidance, for example, relevance to the task. In clump scanning, an attentional window, in which parallel processing of all items occurs, moves systematically between spatially adjacent locations. Importantly, Liesefeld and Müller (2020) argued that priority maps are useful only if that is the more efficient strategy than the alternative in a particular setting. Thus, they suggest that participants may switch between search strategies according to the search task. Using their terms, we propose that our participants were engaging in priority guidance when a valid cue was present, marking the respective location as high priority, and thus, the location of the salient distractor was marked as lower-priority location when such a distractor was present. Following a neutral cue, the difference in priority level for the target and the distractor's locations was somewhat smaller, but priority guidance could still be efficiently used when the task was relatively easy, as in Experiment 1. However, when the target and the surrounding non-targets were more similar, as in our Experiment 2, the difference in priority levels for the target and distractor locations was considerably smaller, making clump scanning the more efficient strategy. Because the locations of the target and the surrounding items in the display are chosen randomly for each trial, the systematic nature of the proposed clump scanning (e.g., clockwise) results in randomly selected items, as we proposed above. One could also argue that the difficult task in Experiment 2 led to a change of strategy also following the valid cue, in which case the cue was not used to boost the priority of the target location but marking the first location to be

searched when clump scanning is initiated. Although we cannot provide clear-cut evidence for the employment of these strategies, our results seem to support the proposed theoretical framework.

4.2 | Top-down guidance of attention reduces the need for distractor suppression

Our cueing manipulation also showed an effect on the P_D in Experiment 1, where salient distractors elicited larger P_D amplitudes when the allocation of attention to the target location was not facilitated by the cue. However, while going in the same direction, the P_D was weak or absent in Experiment 2, suggesting that suppression was not equally engaged in that setting following both cues. Recently, Berggren and Eimer (2020) used a go/no-go task with a central cue that indicated which hemifield should be attended and which should be ignored. They found that the P_D was evident for irrelevant targets on the to-be-ignored side in both singleton- and feature-search tasks, suggesting that salient irrelevant objects triggered active inhibition outside the focus of attention regardless of their relevance. This result was interpreted as evidence for active distractor suppression at the non-cued side. As such, if the non-cued locations in our study were actively suppressed when the target location was cued, we would have expected to see a P_D for salient distractors following a valid cue. The lack of such a finding in the current study suggests that the suppression mechanism that is reflected by the P_D can be made redundant when attention is efficiently allocated elsewhere. This discrepancy suggests different deployment of attention in the settings of the two studies, for example, the use of invalid instead of neutral cues interleaved with valid ones. However, Berggren and Eimer (2020) also found that no P_D was elicited when irrelevant targets in the to-be-ignored hemifield were embedded in a heterogeneous array, leading to the conclusion that salience is necessary for the activation of this type of suppression. Our results suggest that salience was not enough in our paradigm.

The pattern of results found in our study is in accordance with the signal suppression hypothesis (Gaspelin & Luck, 2018a, 2018b; Sawaki & Luck, 2010). According to Sawaki and Luck (2010), the only exception where a salient distractor will not be detected and capture attention, is when attention is strongly focused on a different area of the display. This claim is supported in our study by two findings; first, the absence of the P_D following a valid cue found here suggests that attention was already focused on the target location, preventing the salient distractor from being detected, thus, abolishing the need for it to be suppressed. Additionally, the lack

of a significant P_D in Experiment 2 when there was no top-down guidance of attention also points in the same direction. This is because if the target could not have been easily found in that condition, attention would have been focused on each of the objects in the search array in turn (starting with the salient target if one was present), and thus, there was little chance for suppression to occur. Second, the N2pc found for salient targets following a neutral cue in Experiment 2 suggest that in the absence of top-down attentional guidance, the salient target lead to the allocation of attention to allow processing in that location.

In contrast to previous studies that showed strong suppression resulting in no interference, or even a facilitation effect on performance in distractor-present trials, specifically using small search displays with a small number of items (Gaspelin et al., 2015; Gaspelin & Luck, 2018b; Kerzel & Burra, 2020), in our study performance severely declined when a salient distractor was present, which is the more common outcome (Theeuwes, 2018). Thus, our results suggest that top-down suppression was not perfect in the conditions employed here. One reason for this could be that our participants could not adopt a simple a-priori rejection template, as suggested by some studies (e.g., Arita et al., 2012; Sawaki & Luck, 2010). Because on some trials the target could be the color singleton, a strategy for suppressing all color singletons would not be beneficial (see Kerzel & Burra, 2020). Similarly, as the cardinal color of the search-display changed randomly, no feature-based rejection-template could be adopted (but see Liesefeld et al., 2021). Recently, it was proposed that the inhibition of a salient color singleton is a result of selection history rather than top-down knowledge of the distractor features and may be possible through practice (Gaspelin & Luck, 2019). Presumably, as the salient distractor in our experiments was always a color singleton, such learning through selection history was possible. However, since the color singleton was not uniquely reserved for the salient distractor learning could have been impaired, resulting in faulty suppression.

5 | CONCLUSION

In sum, our behavioral results provide clear evidence for the facilitation of target processing by the pre-cue, and the interference that comes about from the salient distractor. In addition, the results regarding EEG markers of attention (i.e., N2pc, P_D) suggest that task difficulty influences the strategy for utilizing central cues. Thus, our investigation of the combined effect of top-down attentional control via central cues and bottom-up salience gives support to the signal suppression hypothesis and to a search

dichotomy that allows the switch between strategies according to search efficiency in a specific setting.

CONFLICT OF INTEREST

None.

AUTHOR CONTRIBUTIONS

Einat Rashal: Conceptualization; Data curation; Formal analysis; Investigation; Methodology; Visualization; Writing – original draft. **Mehdi Senoussi:** Formal analysis; Software; Writing – review & editing. **Elisa Santandrea:** Conceptualization; Funding acquisition; Methodology; Project administration; Writing – review & editing. **Suliann Ben Hamed:** Conceptualization; Funding acquisition; Methodology; Project administration; Writing – review & editing. **Emiliano Macaluso:** Conceptualization; Funding acquisition; Methodology; Project administration; Writing – review & editing. **Leonardo Chelazzi:** Conceptualization; Funding acquisition; Methodology; Project administration; Writing – review & editing. **Carsten Nico Boehler:** Conceptualization; Funding acquisition; Investigation; Methodology; Project administration; Resources; Supervision; Writing – original draft.

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REFERENCES

- Anderson, B. A. (2016). The attention habit: How reward learning shapes attentional selection. *Annals of the New York Academy of Sciences*, 1369(1), 24–39. <https://doi.org/10.1111/nyas.12957>
- Arita, J. T., Carlisle, N. B., & Woodman, G. F. (2012). Templates for rejection: Configuring attention to ignore task-irrelevant features. *Journal of Experimental Psychology: Human Perception and Performance*, 38(3), 580–584. <https://doi.org/10.1037/a0027885>
- Bacon, W. F., & Egeth, H. E. (1994). Overriding stimulus-driven attentional capture. *Perception & Psychophysics*, 55(5), 485–496. <https://doi.org/10.3758/bf03205306>
- Barras, C., & Kerzel, D. (2016). Active suppression of salient-but-irrelevant stimuli does not underlie resistance to visual interference. *Biological Psychology*, 121(Pt A), 74–83. <https://doi.org/10.1016/j.biopsycho.2016.10.004>
- Barras, C., & Kerzel, D. (2017). Salient-but-irrelevant stimuli cause attentional capture in difficult, but attentional suppression in easy visual search. *Psychophysiology*, 54(12), 1826–1838. <https://doi.org/10.1111/psyp.12962>
- Ben-Shachar, M. S., Lüdtke, D., & Makowski, D. (2020). effectsiz: Estimation of effect size indices and standardized parameters. *Journal of Open Source Software*, 5(56), 2815. <https://doi.org/10.21105/joss.02815>
- Berggren, N., & Eimer, M. (2020). Spatial filtering restricts the attentional window during both singleton and feature-based visual search. *Attention, Perception & Psychophysics*, 82(5), 2360–2378. <https://doi.org/10.3758/s13414-020-01977-5>

- Boehler, C. N., Tsotsos, J. K., Schoenfeld, M. A., Heinze, H.-J., & Hopf, J.-M. (2011). Neural mechanisms of surround attenuation and distractor competition in visual search. *The Journal of Neuroscience: The Official Journal of the Society for Neuroscience*, 31(14), 5213–5224. <https://doi.org/10.1523/JNEUROSCI.6406-10.2011>
- Burra, N., & Kerzel, D. (2013). Attentional capture during visual search is attenuated by target predictability: Evidence from the N2pc, Pd, and topographic segmentation. *Psychophysiology*, 50(5), 422–430. <https://doi.org/10.1111/psyp.12019>
- Chelazzi, L., Perlato, A., Santandrea, E., & Della Libera, C. (2013). Rewards teach visual selective attention. *Vision Research*, 85, 58–72. <https://doi.org/10.1016/j.visres.2012.12.005>
- Cosman, J. D., Lowe, K. A., Zinke, W., Woodman, G. F., & Schall, J. D. (2018). Prefrontal control of visual distraction. *Current Biology*, 28(3), 414–420.e3. <https://doi.org/10.1016/j.cub.2017.12.023>
- Cousineau, D. (2005). Confidence intervals in within-subject designs: A simpler solution to Loftus and Masson's method. *Tutorials in Quantitative Methods for Psychology*, 1(1), 42–45. <https://doi.org/10.20982/tqmp.01.1.p042>
- Di Bello, F., Ben Hadj Hassen, S., Astrand, E., & Ben Hamed, S. (2020). Selection and suppression of visual information in the macaque prefrontal cortex. *BioRxiv*, 2020.03.25.007922. <https://doi.org/10.1101/2020.03.25.007922>
- Donohue, S. E., Bartsch, M. V., Heinze, H.-J., Schoenfeld, M. A., & Hopf, J.-M. (2018). Cortical mechanisms of prioritizing selection for rejection in visual search. *The Journal of Neuroscience: The Official Journal of the Society for Neuroscience*, 38(20), 4738–4748. <https://doi.org/10.1523/JNEUROSCI.2407-17.2018>
- Egeth, H. E., & Yantis, S. (1997). Visual attention: Control, representation, and time course. *Annual Review of Psychology*, 48, 269–297. <https://doi.org/10.1146/annurev.psych.48.1.269>
- Eimer, M. (1996). The N2pc component as an indicator of attentional selectivity. *Electroencephalography and Clinical Neurophysiology*, 99(3), 225–234. [https://doi.org/10.1016/0013-4694\(96\)95711-9](https://doi.org/10.1016/0013-4694(96)95711-9)
- Eimer, M. (2014). The neural basis of attentional control in visual search. *Trends in Cognitive Sciences*, 18(10), 526–535. <https://doi.org/10.1016/j.tics.2014.05.005>
- Feldmann-Wüstefeld, T., Weinberger, M., & Awh, E. (2021). Spatially guided distractor suppression during visual search. *The Journal of Neuroscience: The Official Journal of the Society for Neuroscience*, 41(14), 3180–3191. <https://doi.org/10.1523/JNEUROSCI.2418-20.2021>
- Folk, C. L., Remington, R. W., & Johnston, J. C. (1992). Involuntary covert orienting is contingent on attentional control settings. *Journal of Experimental Psychology: Human Perception and Performance*, 18(4), 1030–1044. <https://doi.org/10.1037/0096-1523.18.4.1030>
- Franconeri, S. L., & Simons, D. J. (2003). Moving and looming stimuli capture attention. *Perception & Psychophysics*, 65(7), 999–1010. <https://doi.org/10.3758/bf03194829>
- Gaspelin, N., Leonard, C. J., & Luck, S. J. (2015). Direct evidence for active suppression of salient-but-irrelevant sensory inputs. *Psychological Science*, 26(11), 1740–1750. <https://doi.org/10.1177/0956797615597913>
- Gaspelin, N., & Luck, S. J. (2018a). Distinguishing among potential mechanisms of singleton suppression. *Journal of Experimental Psychology: Human Perception and Performance*, 44(4), 626–644. <https://doi.org/10.1037/xhp0000484>
- Gaspelin, N., & Luck, S. J. (2018b). Combined electrophysiological and behavioral evidence for the suppression of salient distractors. *Journal of Cognitive Neuroscience*, 30(9), 1265–1280. https://doi.org/10.1162/jocn_a_01279
- Gaspelin, N., & Luck, S. J. (2018c). The role of inhibition in avoiding distraction by salient stimuli. *Trends in Cognitive Sciences*, 22(1), 79–92. <https://doi.org/10.1016/j.tics.2017.11.001>
- Gaspelin, N., & Luck, S. J. (2019). Inhibition as a potential resolution to the attentional capture debate. *Current Opinion in Psychology*, 29, 12–18. <https://doi.org/10.1016/j.copsyc.2018.10.013>
- Gaspelin, N., Ruthruff, E., & Lien, M.-C. (2016). The problem of latent attentional capture: Easy visual search conceals capture by task-irrelevant abrupt onsets. *Journal of Experimental Psychology: Human Perception and Performance*, 42(8), 1104–1120. <https://doi.org/10.1037/xhp0000214>
- Geng, J. J., & Behrmann, M. (2002). Probability cuing of target location facilitates visual search implicitly in normal participants and patients with hemispatial neglect. *Psychological Science*, 13(6), 520–525. <https://doi.org/10.1111/1467-9280.00491>
- Harris, C. R., Millman, K. J., van der Walt, S. J., Gommers, R., Virtanen, P., Cournapeau, D., ... Oliphant, T. E. (2020). Array programming with NumPy. *Nature*, 585(7825), 357–362.
- Hickey, C., Di Lollo, V., & McDonald, J. (2008). Electrophysiological indices of target and distractor processing in visual search. *Journal of Cognitive Neuroscience*, 21, 760–775. <https://doi.org/10.1162/jocn.2009.21039>
- Hickey, C., McDonald, J. J., & Theeuwes, J. (2006). Electrophysiological evidence of the capture of visual attention. *Journal of Cognitive Neuroscience*, 18(4), 604–613. <https://doi.org/10.1162/jocn.2006.18.4.604>
- Hopf, J.-M., Boelmans, K., Schoenfeld, A. M., Heinze, H.-J., & Luck, S. J. (2002). How does attention attenuate target-distractor interference in vision? Evidence from magnetoencephalographic recordings. *Brain Research. Cognitive Brain Research*, 15(1), 17–29. [https://doi.org/10.1016/S0926-6410\(02\)00213-6](https://doi.org/10.1016/S0926-6410(02)00213-6)
- Hopf, J.-M., Luck, S. J., Girelli, M., Hagner, T., Mangun, G. R., Scheich, H., & Heinze, H.-J. (2000). Neural sources of focused attention in visual search. *Cerebral Cortex*, 10(12), 1233–1241. <https://doi.org/10.1093/cercor/10.12.1233>
- Huang, W., Su, Y., Zhen, Y., & Qu, Z. (2016). The role of top-down spatial attention in contingent attentional capture. *Psychophysiology*, 53(5), 650–662. <https://doi.org/10.1111/psyp.12615>
- Jiang, Y. V., Sha, L. Z., & Remington, R. W. (2015). Modulation of spatial attention by goals, statistical learning, and monetary reward. *Attention, Perception, & Psychophysics*, 77(7), 2189–2206. <https://doi.org/10.3758/s13414-015-0952-z>
- Jiang, Y. V., Swallow, K. M., Won, B.-Y., Cistera, J. D., & Rosenbaum, G. M. (2015). Task specificity of attention training: The case of probability cuing. *Attention, Perception, & Psychophysics*, 77(1), 50–66. <https://doi.org/10.3758/s13414-014-0747-7>
- Kerzel, D., & Burra, N. (2020). Capture by context elements, not attentional suppression of distractors, explains the PD with small search displays. *Journal of Cognitive Neuroscience*, 32(6), 1170–1183. https://doi.org/10.1162/jocn_a_01535
- Kiss, M., Grubert, A., & Eimer, M. (2013). Top-down task sets for combined features: Behavioral and electrophysiological evidence for two stages in attentional object selection. *Attention, Perception & Psychophysics*, 75(2), 216–228. <https://doi.org/10.3758/s13414-012-0391-z>

- Kiss, M., Grubert, A., Petersen, A., & Eimer, M. (2012). Attentional capture by salient distractors during visual search is determined by temporal task demands. *Journal of Cognitive Neuroscience*, 24(3), 749–759. https://doi.org/10.1162/jocn_a_00127
- Kiss, M., Van Velzen, J., & Eimer, M. (2008). The N2pc component and its links to attention shifts and spatially selective visual processing. *Psychophysiology*, 45(2), 240–249. <https://doi.org/10.1111/j.1469-8986.2007.00611.x>
- Lawrence, M. (2015). *ez: Easy analysis and visualization of factorial experiments* [R package version 4.3.]. <https://github.com/mike-lawrence/ez> (Original work published 2011).
- Leber, A. B., & Egeth, H. E. (2006). It's under control: Top-down search strategies can override attentional capture. *Psychonomic Bulletin & Review*, 13(1), 132–138. <https://doi.org/10.3758/bf03193824>
- Liesefeld, H. R., Liesefeld, A. M., & Müller, H. J. (2021). Preparatory control against distraction is not feature-based. *Cerebral cortex*, bhab341. <https://doi.org/10.1093/cercor/bhab341>
- Liesefeld, H. R., Liesefeld, A. M., Töllner, T., & Müller, H. J. (2017). Attentional capture in visual search: Capture and post-capture dynamics revealed by EEG. *NeuroImage*, 156, 166–173. <https://doi.org/10.1016/j.neuroimage.2017.05.016>
- Liesefeld, H. R., & Müller, H. J. (2020). A theoretical attempt to revive the serial/parallel-search dichotomy. *Attention, Perception & Psychophysics*, 82(1), 228–245. <https://doi.org/10.3758/s13414-019-01819-z>
- Luck, S. J., & Gaspelin, N. (2017). How to get statistically significant effects in any ERP experiment (and why you shouldn't). *Psychophysiology*, 54(1), 146–157. <https://doi.org/10.1111/psyp.12639>
- Luck, S. J., Girelli, M., McDermott, M. T., & Ford, M. A. (1997). Bridging the gap between monkey neurophysiology and human perception: An ambiguity resolution theory of visual selective attention. *Cognitive Psychology*, 33(1), 64–87. <https://doi.org/10.1006/cogp.1997.0660>
- Luck, S. J., & Hillyard, S. A. (1994). Spatial filtering during visual search: Evidence from human electrophysiology. *Journal of Experimental Psychology: Human Perception and Performance*, 20(5), 1000–1014. <https://doi.org/10.1037/0096-1523.20.5.1000>
- Luck, S. J., Woodman, G. F., & Vogel, E. K. (2000). Event-related potential studies of attention. *Trends in Cognitive Sciences*, 4(11), 432–440. [https://doi.org/10.1016/S1364-6613\(00\)01545-X](https://doi.org/10.1016/S1364-6613(00)01545-X)
- Maris, E., & Oostenveld, R. (2007). Nonparametric statistical testing of EEG-and MEG-data. *Journal of Neuroscience Methods*, 164(1), 177–190. <https://doi.org/10.1016/j.jneumeth.2007.03.024>
- Morey, R. D. (2008). Confidence intervals from normalised data: A correction to Cousineau (2005). *Tutorials in Quantitative Methods for Psychology*, 4(2), 61–64.
- Peirce, J. W. (2007). PsychoPy—Psychophysics software in Python. *Journal of Neuroscience Methods*, 162(1–2), 8–13. <https://doi.org/10.1016/j.jneumeth.2006.11.017>
- Posner, M. I. (1980). Orienting of attention. *The Quarterly Journal of Experimental Psychology*, 32(1), 3–25. <https://doi.org/10.1080/00335558008248231>
- R Core Team. (2016). *R: A language and environment for statistical computing*. <https://www.gbif.org/tool/81287/r-a-language-and-environment-for-statistical-computing>
- Sawaki, R., Geng, J. J., & Luck, S. J. (2012). A common neural mechanism for preventing and terminating the allocation of attention. *The Journal of Neuroscience: The Official Journal of the Society for Neuroscience*, 32(31), 10725–10736. <https://doi.org/10.1523/JNEUROSCI.1864-12.2012>
- Sawaki, R., & Luck, S. J. (2010). Capture versus suppression of attention by salient singletons: Electrophysiological evidence for an automatic attend-to-me signal. *Attention, Perception & Psychophysics*, 72(6), 1455–1470. <https://doi.org/10.3758/APP.72.6.1455>
- Schapiro, A. C., Turk-Browne, N. B., Botvinick, M. M., & Norman, K. A. (2017). Complementary learning systems within the hippocampus: A neural network modelling approach to reconciling episodic memory with statistical learning. *Philosophical Transactions of the Royal Society B: Biological Sciences*, 372(1711), 20160049. <https://doi.org/10.1098/rstb.2016.0049>
- Schevernels, H., Krebs, R. M., Santens, P., Woldorff, M. G., & Boehler, C. N. (2014). Task preparation processes related to reward prediction precede those related to task-difficulty expectation. *NeuroImage*, 84, 639–647. <https://doi.org/10.1016/j.neuroimage.2013.09.039>
- Seiss, E., Kiss, M., & Eimer, M. (2009). Does focused endogenous attention prevent attentional capture in pop-out visual search? *Psychophysiology*, 46(4), 703–717. <https://doi.org/10.1111/j.1469-8986.2009.00827.x>
- Su, Y., Huang, W., Yang, N., Yan, K., Ding, Y., & Qu, Z. (2020). Attentional capture by a color singleton is stronger at spatially relevant than irrelevant locations: Evidence from an ERP study. *Psychophysiology*, 57(10), e13640. <https://doi.org/10.1111/psyp.13640>
- Tan, M., & Wyble, B. (2015). Understanding how visual attention locks on to a location: Toward a computational model of the N2pc component. *Psychophysiology*, 52(2), 199–213. <https://doi.org/10.1111/psyp.12324>
- Theeuwes, J. (1991). Exogenous and endogenous control of attention: The effect of visual onsets and offsets. *Perception & Psychophysics*, 49(1), 83–90. <https://doi.org/10.3758/bf03211619>
- Theeuwes, J. (1992). Perceptual selectivity for color and form. *Perception & Psychophysics*, 51(6), 599–606. <https://doi.org/10.3758/bf03211656>
- Theeuwes, J. (2010). Top-down and bottom-up control of visual selection. *Acta Psychologica*, 135(2), 77–99. <https://doi.org/10.1016/j.actpsy.2010.02.006>
- Theeuwes, J. (2018). Visual selection: Usually fast and automatic; seldom slow and volitional. *Journal of Cognition*, 1(1), 29. <https://doi.org/10.5334/joc.13>
- Töllner, T., Rangelov, D., & Müller, H. J. (2012). How the speed of motor-response decisions, but not focal-attentional selection, differs as a function of task set and target prevalence. *Proceedings of the National Academy of Sciences of the United States of America*, 109(28), E1990–E1999. <https://doi.org/10.1073/pnas.1206382109>
- Treisman, A. M., & Gelade, G. (1980). A feature-integration theory of attention. *Cognitive Psychology*, 12(1), 97–136. [https://doi.org/10.1016/0010-0285\(80\)90005-5](https://doi.org/10.1016/0010-0285(80)90005-5)
- van den Berg, B., Krebs, R. M., Lorist, M. M., & Woldorff, M. G. (2014). Utilization of reward-prospect enhances preparatory attention and reduces stimulus conflict. *Cognitive, Affective & Behavioral Neuroscience*, 14(2), 561–577. <https://doi.org/10.3758/s13415-014-0281-z>
- Yantis, S., & Jonides, J. (1984). Abrupt visual onsets and selective attention: Evidence from visual search. *Journal of Experimental*

Psychology: Human Perception and Performance, 10(5), 601–621. <https://doi.org/10.1037//0096-1523.10.5.601>

Zelinsky, G. J., & Bisley, J. W. (2015). The what, where, and why of priority maps and their interactions with visual working memory. *Annals of the New York Academy of Sciences*, 1339, 154–164. <https://doi.org/10.1111/nyas.12606>

Zivony, A., Allon, A. S., Luria, R., & Lamy, D. (2018). Dissociating between the N2pc and attentional shifting: An attentional blink study. *Neuropsychologia*, 121, 153–163. <https://doi.org/10.1016/j.neuropsychologia.2018.11.003>

SUPPORTING INFORMATION

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FIGURE S1 We ran permutation tests to establish the presence of the N2pc and Pd, since the collapsed-localizer approach could have biased these simple tests to be significant in traditional statistical testing. To allow a direct comparison with the main analysis, the permutation tests were based on the mean amplitudes obtained for each participant in each condition. The permutation procedure consisted of randomly permuting the sign of the difference between ipsi- and contralateral electrodes for each subject independently (Maris & Oostenveld, 2007). After permuting the signs, the average across subjects was calculated, yielding a surrogate ERP component value under the null hypothesis that there was no difference between the potential in the ipsi- and contralateral electrodes in a certain condition. This permutation was repeated 1000 times, yielding a distribution of values constituting an empirically determined estimate of the null distribution. If the actual observed ERP value was smaller or larger than 95% of values in the surrogate distribution, for the N2pc and Pd respectively, then the null hypothesis was rejected. In all plots, the histogram represents the surrogate distribution from the permutation test and the red vertical bar represents the observed ERP value. The first two rows represent Experiment 1, the last two rows represent Experiment 2. The first three columns represent the N2pc for the different lateral-target, no-distractor, and salient-target conditions respectively. The right-most column represents the Pd. (a) Surrogate distribution

and observed N2pc in Experiment 1 neutral cue lateral-target condition ($p < .001$). (b) Surrogate distribution and observed N2pc in Experiment 1 neutral cue no-distractor condition ($p < .001$). (c) Surrogate distribution and observed N2pc in Experiment 1 neutral cue salient-target condition ($p = .01$). (d) Surrogate distribution and observed N2pc in Experiment 1 valid cue lateral-target condition ($p = .63$). (e) Surrogate distribution and observed N2pc in Experiment 1 valid cue no-distractor condition ($p = .54$). (f) Surrogate distribution and observed N2pc in Experiment 1 valid cue salient-target condition ($p = .09$). (g) Surrogate distribution and observed Pd in Experiment 1 neutral cue midline-target condition ($p < .001$). (h) Surrogate distribution and observed Pd in Experiment 1 valid cue midline-target condition ($p = .29$). (i) Surrogate distribution and observed N2pc in Experiment 2 neutral cue Lateral-target condition ($p = .59$). (j) Surrogate distribution and observed N2pc in Experiment 2 neutral cue no-distractor condition ($p = .60$). (k) Surrogate distribution and observed N2pc in Experiment 2 neutral cue salient-target condition ($p = .01$). (l) Surrogate distribution and observed N2pc in Experiment 2 valid cue lateral-target condition ($p = .02$). (m) Surrogate distribution and observed N2pc in Experiment 2 valid cue no-distractor condition ($p < .02$). (n) Surrogate distribution and observed N2pc in Experiment 2 valid cue salient-target condition ($p = .03$). (o) Surrogate distribution and observed Pd in Experiment 2 neutral cue midline-target condition ($p < .001$). (p) Surrogate distribution and observed Pd in Experiment 2 valid cue midline-target condition ($p = .37$)

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